GENERAL SCIENCE GRADE 10: RESPIRATION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.5 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.5: Respiration
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Aerobic and anaerobic respiration in living things – Respiratory substrates and respiratory quotient (RQ) – Factors affecting respiration in living things – Economic importance of anaerobic respiration at home and in industry – Fermentation investigations using yeast cells – Properties of fermented products (acidity) – Project: value-addition products using anaerobic respiration (yoghurt, uji/porridge, biogas)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the meaning of respiration in living things, b) describe aerobic and anaerobic respiration in living things, c) relate the respiratory quotient to the type of substrate and type of respiration in living things, d) explain factors affecting respiration in living things, e) describe economic importance of anaerobic respiration at home and in industry, f) appreciate the significance of respiration in living things.
Core Competencies	– Communication and Collaboration: as the learner discusses with peers the factors affecting energy requirements in human beings, presents findings on aerobic and anaerobic respiration in plenary, and collaborates on fermentation investigations and value-addition projects. – Learning to Learn: as the learner carries out activities to investigate anaerobic respiration in living organisms (fermentation using yeast cells), reflects on experimental results, and adjusts approaches based on evidence gathered.
Core Values	– Integrity: as the learner observes ethical behaviour when working on a project of value addition of food products (such as yoghurt and uji/porridge) that utilises the concept of anaerobic respiration, accurately recording results without falsification. – Peace: as the learner works harmoniously with others in carrying out a project that utilises the concept of anaerobic respiration such as making of yoghurt, porridge, biogas, respecting group roles and others' contributions.
Science & Engineering Practices	– Asking Questions and Defining Problems: as learners formulate testable questions about why dough rises, why uji turns sour, or why a sealed bottle of juice fizzes — anchoring inquiry in observable Kenyan food phenomena before formal instruction. – Planning and Carrying Out Investigations: as learners design and conduct fermentation experiments using yeast cells,

	controlling variables (temperature, sugar concentration, substrate type) and recording observations systematically. – Analysing and Interpreting Data: as learners calculate the Respiratory Quotient (RQ) from experimental or provided data to determine the type of substrate and type of respiration occurring in a living organism. – Constructing Explanations: as learners use evidence from fermentation investigations and RQ calculations to explain the difference between aerobic and anaerobic respiration and relate this to real-world food and energy processes in Kenya. – Engaging in Argument from Evidence: as learners present and defend their conclusions about factors affecting respiration and the economic value of anaerobic respiration in peer discussions and plenary sessions.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: as the learner undertakes a project that utilises the concept of anaerobic respiration in value addition of food products such as milk (yoghurt) and flour/maize (uji/porridge), recognising income-generating potential for Kenyan households and small enterprises. – Environmental Issues: as the learner develops the skill of making biogas as an alternative source of fuel (energy), understanding how anaerobic respiration can reduce dependence on firewood and charcoal, and contribute to environmental sustainability in Kenya.
Career Connections	– Food Scientist / Food Technologist: this sub-strand directly connects to fermentation science underlying production of yoghurt, uji, togwa, and other fermented Kenyan foods — a growing agro-processing industry sector. – Biochemist / Biotechnologist: understanding aerobic and anaerobic pathways, RQ calculations, and enzyme-driven substrate breakdown underpins careers in biochemical research, pharmaceutical production, and industrial biotechnology. – Agricultural Engineer / Biogas Engineer: the economic importance of anaerobic respiration connects to biogas production from animal waste and crop residues, a growing renewable energy field relevant to Kenyan rural communities and Vision 2030. – Nutritionist / Dietitian: knowledge of respiratory substrates (carbohydrates, fats, proteins) and how the body selects and oxidises them connects directly to dietary planning, sports nutrition, and public health practice in Kenya. – Brewer / Dairy Technologist: fermentation principles underlie industrial production of dairy products, non-alcoholic malt drinks, and other fermented beverages, offering entrepreneurship pathways within Kenya's food and beverage industry.
Focus for Lessons	This unit investigates how living organisms — from yeast cells fermenting maize flour into uji, to the muscles of a Kenyan marathon runner pushing through the final kilometre — release energy from food through aerobic and anaerobic respiration. Learners move from concrete, locally familiar observations (rising dough, souring milk, biogas digesters in rural Kenya) toward quantitative understanding of respiratory substrates and the Respiratory Quotient, and then outward to the economic and environmental significance of anaerobic processes. The unit is anchored in hands-on fermentation investigations and a value-addition project, ensuring that scientific understanding is directly connected to entrepreneurship, food security, and environmental sustainability in the Kenyan context.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities? KICD KEY INQUIRY QUESTION: 1. How is aerobic and anaerobic respiration important in a society's economy?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Sealed Thermos of Uji That Keeps Changing" A Form 4 student in Kisumu notices something puzzling during the school's morning break. She pours uji (fermented maize porridge) from a thermos flask into a cup and sees tiny bubbles rising inside the flask. When she smells the remaining uji, it is more sour than it was when she left home two hours ago. The thermos was sealed the entire time — no new ingredients were added, no heat was applied — yet the uji has visibly changed: it bubbles, smells more acidic, and tastes increasingly sour. The student is

	confused: where are the bubbles coming from if the flask was sealed? What is producing the sour taste? Is something alive inside the uji making these changes happen? And why does the same thing NOT happen to fresh maize porridge (uji wa kawaida) that has not been fermented? This puzzling, everyday observation — invisible agents transforming food chemistry inside a sealed container using only the food itself as fuel — is the anchoring phenomenon for the entire unit on respiration. It raises immediate questions about what respiration really means, who or what is doing it, what molecules are being consumed and produced, and why these processes matter for both the body and the economy.
Storyline Thread	Lesson 1 — PREDICT & ANCHOR: Students observe the anchoring phenomenon (sealed thermos of sour, bubbling uji) and make written predictions about what is happening inside and why. Class constructs an initial Driving Question Board (DQB), generating "need-to-know" questions. Teacher introduces the distinction between breathing and cellular respiration to begin resolving confusion. Lesson 2 — OBSERVE & EXPLORE (Aerobic Respiration): Learners discuss the meaning of respiration in living things and use print/digital resources to investigate aerobic respiration — word equation, reactants, products, and sites of reaction. Connecting to the sprinter/marathon runner supporting phenomenon, learners explain why continuous oxygen supply matters for sustained energy release, and map aerobic respiration onto human dietary substrates (ugali → glucose → ATP). Lesson 3 — OBSERVE & INVESTIGATE (Anaerobic Respiration in Yeast): Learners carry out a guided fermentation investigation — yeast + glucose solution in a sealed flask with a lime-water indicator tube — observing CO_2 production and comparing to a no-yeast control. Results are recorded and shared in plenary. The anchoring phenomenon (uji bubbles) is revisited: learners now explain what is producing the bubbles in the thermos. Lesson 4 — EXPLAIN (Anaerobic Respiration in Animals & Plants; Products & Equations): Learners describe anaerobic respiration in animals (lactic acid pathway — connecting to the burning thigh muscles of the Nairobi sprinter) and in yeast/plants (ethanol + CO_2 pathway). Word and chemical equations are written. Learners test fermented product acidity using universal indicator, connecting pH change to the sour taste of uji and mala. Lesson 5 — EXPLAIN (Respiratory Substrates & Respiratory Quotient): Learners use print/digital resources to distinguish respiratory substrates (carbohydrates, fats, proteins) and are guided to calculate the RQ using provided data sets (e.g., RQ for glucose = 1, for fat ≈ 0.7, for protein ≈ 0.8). Learners relate RQ values to the type of substrate being respired and the type of respiration occurring, applying calculations to Kenyan dietary contexts (maize-based ugali vs. nyama choma/meat). Lesson 6 — EXPLAIN (Factors Affecting Respiration): Learners discuss and investigate factors affecting the rate of respiration — temperature, oxygen availability, substrate concentration, surface area (germinating beans vs. dry seeds). Results are presented in peer groups and connected back to the anchoring phenomenon: why does the uji ferment faster on a hot Mombasa afternoon than in a cool Kericho highland morning? Lesson 7 — DRIVING QUESTION BOARD REVISION & ECONOMIC IMPORTANCE: Learners revisit and update the DQB, crossing off questions now answered and identifying remaining gaps. They then research and present (using digital/print media) the economic importance of anaerobic respiration — yoghurt/mala production, uji/togwa fermentation, biogas from the Rift Valley dairy farm, silage making, bread and mandazi leavening — connecting science directly to entrepreneurship and Kenya's Vision 2030 agro-processing goals. CAUTION on brewing of alcoholic substances is explicitly discussed. Lesson 8 — MODEL BUILDING & PROJECT SHOWCASE: Learner groups present their value-addition projects (yoghurt,

	uji/porridge fermentation, or biogas model) and revise their cumulative explanatory model of respiration. Each group submits a final annotated model diagram connecting the anchoring phenomenon (sealed uji thermos) to aerobic/anaerobic pathways, RQ, factors affecting respiration, and economic applications. Peer and teacher assessment using the KICD rubric on RQ, factors, and experimental procedure.
Supporting Phenomena	– Rising Chapati Dough: a Nairobi street vendor's dough ball, left covered for 30 minutes after adding a pinch of instant yeast and sugar, visibly doubles in size and develops a slightly alcoholic smell — yet no air was pumped in. Where did the gas come from, and what process produced it? – The Sprinter vs. the Marathon Runner: during the Nairobi Schools Athletics Championships, the 100 m sprinter crosses the finish line gasping and feeling a burning sensation in her thighs, while the 10 km runner breathes harder but steadily throughout. Why do their muscles respond so differently even though both are "running"? – Biogas Digester at a Rift Valley Dairy Farm: a farmer near Nakuru feeds cow dung into a sealed underground tank and, two weeks later, has enough gas through the outlet pipe to cook three meals a day for his family. Nothing was burned — so where does the cooking gas come from, and what is happening inside the sealed tank? – Fresh Milk vs. Mala (Fermented Milk): fresh milk left at room temperature in Mombasa's heat turns solid, smells sour, and separates into curds and whey within hours — but a dairy factory deliberately uses this same process to produce mala and yoghurt that people pay for. What organisms and chemical processes are responsible, and how is the outcome controlled?

LESSON 1 (80 minutes (2 × 40-minute lessons)): The Sealed Thermos of Uji That Keeps Changing — Predict & Anchor	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon, elicit prior conceptions about respiration, and co-construct the class Driving Question Board (DQB) so that every subsequent lesson has a shared investigative purpose.
Knowledge	Learners will distinguish between breathing (ventilation) and cellular respiration; identify that fermented uji undergoes chemical change inside a sealed container; and recognise that invisible living agents (microorganisms) can transform food chemistry by consuming substrates and releasing products.
Skills	Learners will practise the NGSS Science & Engineering Practice of Asking Questions and Defining Problems by generating testable, need-to-know questions; and the Practice of Constructing Explanations by writing initial evidence-based predictions.
Attitudes	Learners will appreciate the significance of respiration in living things (KICD SLO f) and demonstrate integrity by recording honest predictions even when uncertain.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	This is the opening lesson of the unit. It anchors every subsequent lesson to a single puzzling, locally relevant observation — sealed, fermented uji that bubbles and sours without any external input. By predicting before instruction, learners surface prior knowledge and misconceptions that the teacher will use to sequence later lessons. The DQB created here will be revisited and updated in every subsequent lesson, giving learners ownership of the unit's driving questions.
Safety Notes	No chemicals are handled in this lesson. If a real thermos of fermented uji is brought in as a demonstration, the teacher must ensure the flask is opened away from learners' faces to avoid any unexpected pressure release. Learners with known allergies to fermented foods should not taste the uji. The teacher must remind learners that the KICD curriculum expressly prohibits the brewing of alcoholic substances — fermentation demonstrations must remain non-alcoholic food contexts such as porridge and yoghurt.

B. LESSON OVERVIEW
Learners encounter the anchoring phenomenon for the first time: a sealed thermos of fermented uji (fermented maize porridge) brought from home to school in Kisumu has changed — it now bubbles and smells more sour — even though it was sealed the entire time with no heat or new ingredients added. Working individually and then in groups, learners write and share predictions about what is happening inside the flask. The class then pools their questions onto a Driving Question Board. In the final phase, the teacher introduces the critical distinction between breathing and cellular respiration, giving learners just enough vocabulary to begin making sense of the phenomenon without resolving it — preserving productive curiosity for the lessons ahead.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — PREDICT (20	Learners are seated in groups of 4–5. Each learner receives a blank 'Predict Card' (half-sheet of paper)	1. SETUP (2 min): Place the sealed thermos on the demonstration table before	INDIVIDUAL PREDICTION WRITING — This strategy (aligned to NGSS SEP 1: Asking Questions	OBSERVABLE INDICATOR: The teacher reads Predict Cards during the 5-minute independent writing

minutes) 📹 **VIDEO:** [Cellular respiration introduction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	divided into three columns: WHAT I OBSERVE / WHAT I THINK IS HAPPENING / WHAT EVIDENCE WOULD CHANGE MY MIND. The teacher places a sealed thermos flask at the front of the room and narrates the Kisumu scenario (a Form 4 student notices bubbling, increased sourness in her morning uji) without opening the flask yet. Learners silently and independently write their predictions for 5 minutes. They then share with their table group, listening for similarities and differences in their ideas. Each group appoints a spokesperson to share one key prediction with the class. Learners are not told whether any prediction is correct — all ideas are treated as equally valid hypotheses at this stage.	learners enter. When learners are seated, say: 'Do not touch the flask yet. Just look at it. Think about what might be inside it right now.' Allow 30 seconds of silent observation. 2. NARRATE THE PHENOMENON (3 min): Read the Kisumu scenario aloud slowly and deliberately: 'A Form 4 student at a school near Lake Victoria poured fermented uji — uji wa kimea — from her thermos at morning break. She noticed tiny bubbles rising inside the flask. The uji smelled more sour than when she left home two hours earlier. The flask had been sealed the entire time. No heat. No new ingredients. Yet the uji had changed.' Pause after the final sentence for 10 seconds of WAIT TIME. 3. PROMPT INDIVIDUAL PREDICTION (5 min): Distribute Predict Cards. Say: 'On your card, write what YOU think is happening inside that sealed thermos. You have five minutes. Write in silence. There are no wrong answers — I want your real thinking, not the textbook answer.' Circulate and read cards silently; do not correct anything. 4. GROUP SHARE (5 min): Say: 'Share your prediction with your group. Listen carefully — note where your ideas agree and where they differ.' Circulate; listen for the following predicted misconceptions to address later: (a) 'heat made it sour', (b) 'the flask was not properly sealed', (c) 'it is just breathing'. 5. CLASS COLD-CALL (5 min): Cold-call exactly 5 groups (not volunteers) using a name-stick draw. For each group say: 'Group	and Defining Problems) activates prior knowledge and makes thinking visible before instruction. By writing predictions before seeing data, learners commit to an initial mental model that the rest of the unit will progressively revise. The three-column Predict Card scaffold ensures learners also think about what evidence would change their minds — building epistemic habits essential to scientific reasoning.	phase. Look for: (1) Does the learner identify any agent or process causing the change, or do they attribute it entirely to physical causes (heat, pressure)? (2) Does any learner spontaneously mention living organisms, microbes, yeast, or fermentation? (3) Does any learner use the word 'respiration' — and if so, do they mean breathing or a chemical process? These three markers inform which misconceptions to address in Phase 3. Collect all Predict Cards at the end of the lesson to compare with exit reflections.

		[X], in one sentence — what do you think is inside that flask right now that was NOT there this morning?' After each response, say: 'Thank you. Hold that idea.' Do NOT affirm or correct. Write each group's key phrase on the board in a column labelled OUR PREDICTIONS. 6. BRIDGE QUESTION: After all 5 groups have shared, ask the whole class: 'Raise your hand if your group mentioned something ALIVE inside the flask.' Count raised hands aloud and note the number on the board.		
Phase 2 — OBSERVE (20 minutes) 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher opens the thermos flask and invites structured sensory observation (sight and smell only — no tasting until safety has been confirmed). Learners record observations in their science journals using the sentence stems: 'I can see…', 'I can smell…', 'I notice that compared to fresh uji…'. The teacher then plays a short (3–4 minute) video clip showing side-by-side time-lapse footage of sealed fermented uji (bubbling) versus sealed fresh maize porridge (no change). Learners add observations from the video to their journals. The teacher then shows a second visual: two clear jars on the demonstration table — one containing fermented uji and one containing fresh uji wa kawaida — both sealed. Learners compare the two jars visually and record at least three differences. The group then constructs a shared 'Observation Table' on a piece of manila paper, pooling individual observations, and posts it on the class wall.	1. OPEN THE FLASK (3 min): Say: 'I am going to open the flask now. I want complete silence. Observe everything — colour, sound, smell coming from this direction (gesture toward open window side of room). Do NOT lean over the flask.' Open the flask slowly. Allow 10 seconds of silent observation (WAIT TIME). Ask: 'What did you notice? Hands down — I will cold-call three people.' Cold-call 3 learners by name. Write their exact observations on the board under EVIDENCE FROM THE FLASK. 2. VIDEO OBSERVATION (4 min): Say: 'We will now watch a short clip. As you watch, write in your science journal — do not discuss yet.' Play the video. After the video, say: 'In 30 seconds, write the single most surprising thing you observed.' Allow 30 seconds of WAIT TIME and silent writing. 3. TWO-JAR COMPARISON (5 min): Place the two sealed jars (fermented vs. fresh uji) on the demonstration table. Say: 'Without touching the jars, identify at least	STRUCTURED SENSORY OBSERVATION WITH SENTENCE STEMS — This strategy scaffolds NGSS SEP 3 (Planning and Carrying Out Investigations) at the observational level. Sentence stems ('I can see…', 'I notice that compared to…') prevent learners from jumping to explanation before they have fully described the evidence. The two-jar comparison forces learners to use a control (fresh uji) alongside the variable condition (fermented uji), building the habit of controlled comparison that will be formalised in later experimental lessons.	OBSERVABLE INDICATOR: Circulate during the two-jar comparison and the group Observation Table activity. A learner who is 'approaching expectations' will list only surface-level differences (colour, smell). A learner who 'meets expectations' will also note the presence of bubbles and the absence of change in the fresh sample. A learner who 'exceeds expectations' will begin to hypothesise a cause in their observation notes (e.g., 'bubbles suggest a gas is being produced'). Use these distinctions to decide which groups to cold-call during Phase 3.

		three observable differences between these two samples. Write them down.' Allow 3 minutes. Cold-call 4 learners (different from the first cold-call) to share one observation each. Record on the board under FERMENTED UJI vs. FRESH UJI. 4. GROUP OBSERVATION TABLE (5 min): Say: 'Your group has three minutes to create a shared Observation Table on your manila paper. Use two columns: FERMENTED UJI and FRESH UJI WA KAWAIDA. List every difference your group noticed.' Circulate and prompt groups who list fewer than four differences with: 'Think about smell, appearance, what we heard when the flask was opened, and what the video showed about bubbles.' Post tables on the class wall. 5. CONNECT BACK TO PHENOMENON: Say: 'Everything we just observed happened inside a sealed container. No new ingredients. No heat. The question we must now sit with is: what process — what invisible chemistry — could produce all of these changes using only what was already in the flask?'		
Phase 3 — EXPLAIN (20 minutes) 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	The teacher introduces the critical vocabulary distinction between breathing (ventilation) and cellular respiration using a 'Two Things That Look the Same But Are Different' anchor chart. Learners receive a structured note-frame (T-chart) and fill it in during a guided mini-lecture. The teacher uses two Kenya-relevant analogies: (1) a Kisumu fish market worker's chest rising and falling (breathing/ventilation) versus (2)	1. INTRODUCE THE T-CHART (5 min): Draw a T-chart on the board: BREATHING (UPUMUAJI) | CELLULAR RESPIRATION (UPUMUAJI WA SELI). Say: 'These two terms are often confused. By the end of this unit, you will know the difference precisely. Today we start with the big picture.' Fill in the chart with learners contributing: Breathing = movement of air in and out of lungs / whole-body process /	PREDICT–REVISE–ANNOTATE (PRA) — Learners return to their own written predictions and revise them using newly acquired vocabulary, without erasing original thinking. This strategy makes conceptual change visible: the original colour represents prior knowledge and the revision colour represents growing understanding. PRA is aligned to NGSS SEP 6 (Constructing Explanations) and directly supports the KICD	OBSERVABLE INDICATOR: Review the revised Predict Cards (both colours). A learner 'below expectations' will not use any new vocabulary in their revision. A learner 'approaching expectations' will use 'cellular respiration' but apply it only to human bodies, not to the yeast in the uji. A learner 'meeting expectations' will correctly connect cellular respiration to the microorganisms inside the flask and identify at least one product

[Cellular Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the muscles in a Kenyan marathon runner's legs burning glucose during the Lewa Safari Marathon (cellular respiration). Learners then return to their original Predict Cards and annotate them in a different colour, revising or confirming their initial prediction using the new vocabulary. Groups share revised predictions and the teacher records any new vocabulary learners use correctly on the UNIT WORD WALL. The phase ends with the teacher posing the bridging question that connects this distinction back to the phenomenon: 'If the thermos was sealed — no air coming in — what does that tell us about whether the process inside needs oxygen?'	involves lungs, diaphragm / you can see and feel it. Cellular respiration = chemical reactions inside every cell / releases energy from food / produces CO_2 and water / you cannot see it with the naked eye / happens in yeast cells, muscle cells, ALL living cells. 2. KENYAN ANALOGIES (3 min): Say: 'Think of a mama fua in Mombasa — you can watch her chest rise and fall. That is breathing. Now think of Eliud Kipchoge's leg muscles at the 40th kilometre of a marathon — those muscle cells are burning glucose to release energy. That invisible chemical process in each muscle cell is cellular respiration. The two are connected but they are NOT the same thing.' Ask: 'Which of these two processes could happen inside a sealed thermos flask?' Allow 10 seconds of WAIT TIME. Cold-call 3 learners. 3. PREDICT CARD REVISION (5 min): Distribute a coloured pen (different from the original) to each learner. Say: 'Look at your Predict Card. Without crossing anything out — in this new colour — add, revise, or confirm your prediction using at least one of the new terms: cellular respiration, substrate, product, microorganism.' Allow 4 minutes of silent individual work. 4. GROUP SHARE OF REVISED PREDICTIONS (4 min): Cold-call 3 groups (different from earlier). For each group ask: 'What did you change, and what new word did you use?' Record any correct use of new vocabulary on the UNIT WORD WALL (a piece of manila paper posted on the wall with the heading WORDS WE ARE	outcome of appreciating the significance of respiration in living things, because learners must connect the abstract term to the concrete phenomenon they observed.	(gas/CO_2 or acid). A learner 'exceeding expectations' will also note that the sealed container implies limited or no oxygen, which introduces the aerobic/anaerobic distinction the next lesson will formalise.

		BUILDING). 5. BRIDGING QUESTION (3 min): Say: 'Here is the question that will drive our next three lessons. The thermos was completely sealed. If cellular respiration was happening inside — who was doing it? And if there was no fresh air coming in, does that mean oxygen was available or not? Turn and talk to your partner for 60 seconds. Then I will cold-call two pairs.' After responses, say: 'We do not have the full answer yet. That is exactly why we are going to investigate. Hold that question.'		
Phase 4 — DRIVING QUESTION BOARD (DQB) Creation (15 minutes) 📹 **VIDEO:** [Cellular respiration introduction] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Cellular Respiration (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner receives three sticky notes (or small cut paper squares). They write one 'need-to-know' question per sticky note — questions that, if answered, would help them explain the phenomenon. Learners post their questions onto a large manila-paper DQB under one of three pre-printed column headers: WHAT IS HAPPENING? / WHO IS DOING IT? / WHY DOES IT MATTER? The class reads the board together, and the teacher facilitates a brief clustering activity: groups of similar questions are circled and given a cluster label. The teacher then reveals the unit's Driving Question — 'How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?' — and posts it at the top of the DQB. Learners copy the DQB structure (question clusters and their own question) into their science journals. The DQB remains posted on the	1. DISTRIBUTE STICKY NOTES (1 min): Say: 'You have three sticky notes. Write one question per note — questions YOU genuinely want answered to explain what is happening in the thermos. Not questions you already know the answer to. Real questions.' 2. SILENT WRITING (3 min): Allow 3 minutes of silent individual question writing. Circulate and prompt learners who are stuck: 'Think about what you saw — the bubbles. Where do they come from? What is making the sourness? Why does the same thing not happen to fresh uji?' 3. POST AND CLUSTER (5 min): Invite learners to post their sticky notes under the three column headers. Once all notes are posted, say: 'I am going to read some of these aloud. As I do, raise your hand if your question is similar.' Read 8–10 notes aloud and facilitate clustering. Assign cluster labels co-created with the class, e.g.: 'WHO IS LIVING IN THE UJI?' / 'WHAT GAS IS	DRIVING QUESTION BOARD (DQB) CO-CONSTRUCTION — The DQB is a public, persistent record of the class's collective curiosity. By physically posting questions under thematic clusters, learners externalise their thinking in a way that makes the unit's intellectual journey visible. The DQB is not a decorative poster — it is a formative tool: questions are ticked, moved, or revised as understanding grows. This strategy is aligned to NGSS SEP 1 (Asking Questions) and supports the KICD core competency of Communication and Collaboration, as learners must negotiate where questions belong and what the clusters mean.	OBSERVABLE INDICATOR: Count the number of questions posted under each column. If the majority of questions fall under 'WHAT IS HAPPENING?' with few under 'WHY DOES IT MATTER?', this signals that learners are not yet connecting the phenomenon to economic or societal implications — the teacher should plan a targeted prompt at the start of Lesson 2 to bridge this gap. Also check whether any learner spontaneously wrote questions about oxygen, anaerobic processes, or ATP — these learners may have prior knowledge from Junior School Integrated Science and can be positioned as peer-teaching resources in later lessons.

	classroom wall for the duration of the unit.	FORMING THE BUBBLES?' / 'WHY IS IT MORE SOUR?' / 'DOES THIS NEED OXYGEN?' / 'HOW IS THIS USEFUL IN REAL LIFE?' 4. REVEAL THE DRIVING QUESTION (2 min): Say: 'All of your questions are excellent. They are exactly the questions scientists ask. Here is the question that ties them all together for this unit.' Read the Driving Question aloud. Post it at the top of the DQB. Say: 'By the end of this unit, every cluster of questions on this board will have been answered. We will come back to this board every lesson to tick off what we now understand and add new questions as we learn more.' 5. JOURNAL COPY (2 min): Say: 'Copy the DQB structure — the three column headers, your question clusters, and the Driving Question — into your science journal. Date it. This is Day One of our investigation.' 6. POINT TO KICD KEY INQUIRY QUESTION: Say: 'Notice the last cluster — HOW IS THIS USEFUL IN REAL LIFE? The KICD asks us: How is aerobic and anaerobic respiration important in a society's economy? That question — and your questions about the uji — are the same question wearing different clothes.'		
Phase 5 — MODEL BUILDING (5 minutes) 📹 **VIDEO:** [Cellular respiration introduction] Source: Khan Academy (English	As a closing activity, each learner draws a simple Initial Model in their science journal. The model must answer: 'What do you currently think is happening inside the sealed thermos?' The model should include at least: the thermos flask (drawn as a container), some representation of	1. FRAME THE MODEL (1 min): Say: 'Scientists build models to represent their current best thinking. Your model does not need to be correct — it needs to be honest. Draw what YOU currently think is happening inside that thermos. You have four minutes. Use arrows, labels, and	INITIAL MODEL DRAWING — Drawing an initial model before instruction is a cornerstone of the NGSS SEP 2 (Developing and Using Models). It forces learners to commit their current mental model to paper, making implicit assumptions visible to both the learner and the teacher. The	OBSERVABLE INDICATOR: Scan models as you circulate. Look for three levels: (1) BELOW EXPECTATIONS — learner draws only the flask with no internal content or process; (2) APPROACHING EXPECTATIONS — learner draws some internal content (liquid) but no agent or

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	what is inside, arrows showing any inputs or outputs the learner thinks exist, and a label for any agent or process they believe is responsible. Learners are told explicitly: 'This model will be WRONG in places — that is fine. We will revise it at the end of every lesson until it is complete. On the last day of this unit, you will compare your Lesson 1 model to your final model and reflect on how your thinking changed.' Learners label their model 'INITIAL MODEL — Lesson 1' and date it.	any diagrams that help.' 2. SILENT DRAWING (4 min): Allow 4 minutes of uninterrupted silent drawing. Circulate and observe without commenting on correctness. If a learner draws nothing, prompt: 'Just draw the thermos. Now draw what you think is inside it. Now draw one arrow — is anything going in or coming out?' 3. DO NOT COLLECT: Tell learners to keep this model in their journals. Say: 'Guard this page. It is evidence of your thinking today. It is the starting point of your scientific journey in this unit.' 4. CLOSE WITH PHENOMENON CALLBACK: Say: 'Tomorrow we will begin to investigate what is actually living inside fermented uji — and why it needs to eat to survive. The question from today that should stay with you overnight is: If the thermos was completely sealed, where did the gas for those bubbles come from?'	explicit instruction that the model will be revised — not replaced — establishes the epistemological norm that in science, changing your mind based on evidence is a strength, not a failure. This is especially important in the Kenyan classroom context, where learners are often socialised to seek the 'right answer' rather than to track the evolution of their own understanding.	process; (3) MEETS EXPECTATIONS — learner draws an agent (however labelled — 'germs', 'bacteria', 'something alive') and at least one product (bubbles, gas, sourness); (4) EXCEEDS EXPECTATIONS — learner draws a process with inputs (food/sugar in the uji) and outputs (gas, acid), with an organism as the agent, and may label the process as 'fermentation' or 'respiration'. Photograph or mentally note 2–3 models across these levels to use as discussion anchors at the start of Lesson 2.

D. TEACHER REFLECTION

After this lesson, ask yourself the following questions and record your answers in your professional journal before planning Lesson 2:

1. PHENOMENON ENGAGEMENT: Did all learners — including those who rarely participate — engage with the Predict Card? If not, what barrier existed (language, confidence, relevance) and how will you lower it in Lesson 2?

2. MISCONCEPTION MAPPING: What were the two or three most common misconceptions on the Predict Cards? (Common ones: 'the heat made it sour', 'the flask was leaking', 'it is the same as breathing'.) Plan a specific moment in Lesson 2 to surface and address each one using evidence from the fermentation investigation.

3. DQB QUALITY: Did learners' DQB questions point toward the right level of inquiry? If most questions are very surface-level ('what is in the uji?') with no questions about molecules, energy, or economic applications, plan a DQB prompt at the start of Lesson 2: 'Let us add a new column — WHAT MOLECULES ARE INVOLVED?'

4. BREATHING vs. CELLULAR RESPIRATION: Which learners still appear to conflate the two concepts after the T-chart activity? Identify them by name. In Lesson 2, assign these learners a specific observation role during the yeast fermentation experiment that requires them to distinguish between gas production (cellular respiration) and any physical movement (not present in yeast).

5. KENYAN CONTEXT RESONANCE: Did the uji phenomenon feel authentic and engaging to your specific learners? If most learners in your school come from communities where uji wa kimea is not commonly consumed, identify an equivalent locally fermented food (busaa mash before fermentation is complete, fermented milk/mursik preparation in Nandi, togwa in the Coast region) and adapt the phenomenon description for Lesson 2's revisit.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A sealed thermos flask of fermented uji (uji wa kimea) brought from home to school in Kisumu produced visible bubbles and became increasingly sour over two hours, despite being completely sealed with no heat applied and no new ingredients added. When opened, a faint smell was noticeable. A side-by-side comparison with fresh uji wa kawaida (un-fermented maize porridge) in a sealed jar showed no such changes in the fresh sample.
What did I learn?	Breathing (upumuaji) and cellular respiration (upumuaji wa seli) are two different processes. Breathing is the physical movement of air in and out of the lungs — visible and whole-body. Cellular respiration is a chemical process that occurs inside every living cell, releasing energy from food (substrate) and producing carbon dioxide and other products. The changes in the sealed uji are caused by cellular respiration carried out by living microorganisms already present in the fermented porridge — not by heat or external forces.
How does this explain the phenomenon?	The bubbles and increasing sourness in the sealed thermos of fermented uji are produced by microorganisms (likely yeast and bacteria) carrying out cellular respiration inside the flask. These organisms break down sugars in the uji to release energy, producing carbon dioxide gas (the bubbles) and acids (the sour taste) as by-products. Because the flask is sealed and the process still occurs, we begin to ask: does this type of cellular respiration require oxygen, or can it happen without it? This question — aerobic versus anaerobic respiration — is what the rest of the unit will investigate and explain.

LESSON 2 (40 minutes): Observe and Explore: Aerobic Respiration and the Energy in Our Food	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will investigate the process of aerobic respiration by examining its reactants, products, word equation, and site of reaction, connecting this to how cells in the human body release energy from Kenyan dietary staples such as ugali and githeri.
Knowledge	Learners will be able to state the word equation for aerobic respiration (glucose + oxygen yields carbon dioxide + water + energy/ATP), identify the reactants and products, name the mitochondria as the primary site, and explain why a continuous oxygen supply is essential for sustained energy release in active cells.
Skills	Learners will practise reading and interpreting scientific text and diagrams, constructing a labelled word-equation diagram, comparing aerobic respiration to breathing using evidence from resources, and communicating findings verbally and in written form to peers.
Attitudes	Learners will develop curiosity about invisible chemical processes happening inside their own bodies, appreciate the connection between everyday food choices and cellular energy, and demonstrate collaborative engagement during resource-based investigation.
Key Inquiry Question	How is aerobic respiration important in sustaining the energy needs of living organisms, and how does this connect to what people in Kenya eat and how athletes perform?
Purpose in Storyline	Lesson 2 provides the first half of the explanation for the anchoring phenomenon by establishing what aerobic respiration is and why oxygen matters. Learners now understand that living organisms release energy chemically from food — a foundation needed before Lesson 3 shifts to anaerobic respiration and explains the bubbles and sourness in the sealed uji thermos.
Safety Notes	

B. LESSON OVERVIEW
Learners begin by revisiting their Lesson 1 predictions about the sealed uji thermos and identifying what new information they need. They then use print or digital resources to investigate aerobic respiration in pairs, focusing on the word equation, reactants, products, and site of reaction. A supporting phenomenon — a Nairobi sprinter who collapses when she cannot breathe enough oxygen — is introduced to anchor why continuous oxygen supply matters. Learners map the journey from ugali (starch) through glucose to ATP, and by the end of the lesson they can articulate what aerobic respiration does, where it happens, and why it is not yet sufficient to explain the bubbling, sealed uji. This productive tension drives curiosity toward Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Cellular respiration introduction] Source: Khan	Learners open their science journals and re-read the prediction they wrote in Lesson 1 about what is happening inside the sealed uji thermos. They then respond individually in writing to a focused	Teacher writes the prompt on the board before learners arrive. As learners write silently, teacher circulates and places a small sticky note on journals that contain the word 'oxygen' or 'gas' to	Think-Pair-Share activates prior knowledge from Lesson 1 and from everyday experience of breathing, creating a bridge to the new concept of aerobic respiration. The written individual response	Teacher scans journal entries during circulation to identify whether learners are linking energy release to chemical processes or only to physical breathing. This informs pacing: if

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	prompt on the board: 'In Lesson 1 we said something alive inside the uji is releasing energy. How do you think a living cell gets energy out of food? What ingredients do you think the cell needs, and what waste does it produce?' After 2 minutes of silent writing, learners share predictions with a partner using a Think-Pair-Share protocol before two or three pairs share with the whole class.	acknowledge prior knowledge without correcting yet. Teacher then facilitates sharing: 'I am hearing two big ideas — some of you think the cell needs oxygen and some of you are not sure. Both are useful starting points. Let us investigate.' WAIT TIME: After asking 'What ingredients do you think the cell needs?', teacher counts silently to 8 before taking responses, signalling that thinking time is expected and valued.	ensures every learner commits to a prediction before social influence shapes thinking.	most learners conflate breathing with cellular respiration, teacher spends an extra 60 seconds re-reading the Lesson 1 summary statement before moving to the Observe Phase.
Observe Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs. Each pair receives either a printed resource card (a one-page illustrated text on aerobic respiration adapted from KICD Grade 10 materials, including a simple diagram of a mitochondrion and the word equation) or accesses a vetted digital resource on a tablet or phone. Their task, written on a task card, is to find and record four things in their journals: (1) the word equation for aerobic respiration, (2) the names and descriptions of the reactants, (3) the names and descriptions of the products, and (4) the name and location of the organelle where most aerobic respiration occurs. After 7 minutes of paired investigation, teacher introduces the supporting phenomenon verbally: 'Amina Waweru is a 400-metre sprinter training at Kasarani Stadium in Nairobi. During a race, her leg muscles need enormous amounts of energy very quickly. Her coach notices that when she runs at high altitude in Eldoret, she sometimes collapses near the finish line. Her blood oxygen levels are lower at altitude.' Teacher asks: 'Using what you just read, why would lower oxygen levels	Teacher distributes resource cards and task cards simultaneously to save time. During the 7-minute investigation, teacher visits at least four pairs, asking probing questions: 'You have written down carbon dioxide — where does that carbon actually come from originally?' and 'Your card shows a diagram of the mitochondrion — which part of that diagram connects to what is happening inside a cell in Amina's thigh muscle right now?' When introducing Amina, teacher uses a simple sketch on the board: a stick figure runner, a pair of lungs with an arrow to a muscle cell, and a question mark over the cell. WAIT TIME: After asking 'Why would lower oxygen levels cause Amina to struggle?', teacher waits 10 seconds and explicitly says 'I will wait — this is a thinking question, not a recall question.'	The resource-based paired investigation models scientific reading as a skill — learners are not receiving information passively but locating and recording specific evidence. The Amina/Kasarani supporting phenomenon immediately contextualises aerobic respiration in a vivid, locally grounded scenario, making the abstract word equation feel purposeful. The altitude context also plants a productive seed: what happens when oxygen is not enough — a question that Lesson 3 will answer through anaerobic respiration.	Teacher listens to pair discussions about Amina to assess whether learners can apply the word equation causally (less oxygen means less aerobic respiration means less ATP means less energy for muscle contraction) or whether they are still thinking at the level of 'she cannot breathe.' This diagnostic gap informs the Explain Phase emphasis.

	cause Amina to struggle to sustain her sprint?' Learners discuss in pairs for 2 minutes before sharing.			
Explain Phase 📹 **VIDEO:** [Cellular respiration introduction](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. Teacher draws a large flow diagram on the board titled 'From Ugali to ATP' with five labelled boxes: (1) Ugali/Githeri eaten, (2) Starch digested to glucose, (3) Glucose enters cell, (4) Glucose + Oxygen reacts in mitochondria, (5) Carbon dioxide + Water + ATP released. Learners copy this into their journals and annotate each arrow with one sentence explaining the transformation. Teacher then writes the word equation formally: Glucose + Oxygen yields Carbon Dioxide + Water + Energy (ATP). Learners are asked to underline every word that connects back to either the uji thermos phenomenon or to Amina the sprinter. Teacher facilitates a whole-class discussion: 'We now have a clear picture of aerobic respiration. But look back at the uji thermos — it was SEALED. The thermos was closed with no fresh oxygen entering. If aerobic respiration needs oxygen, and the flask is sealed, does aerobic respiration fully explain the bubbles and the sour taste?' Learners discuss and record their current best explanation and remaining questions in their journals.	When drawing the flow diagram, teacher verbalises each step and pauses at step 4 to ask: 'Where exactly in the cell does this happen? I heard the word mitochondria from several pairs — let us make sure everyone has that.' Teacher uses deliberate think-aloud modelling when annotating the first arrow: 'I am writing: starch is a polysaccharide and digestion breaks it into individual glucose molecules — this is the substrate that will fuel the cell.' For the underline activity, teacher says: 'I want to see your thinking — circle the products that you think might be responsible for the changes in the thermos, and put a question mark next to anything you are still unsure about.' WAIT TIME: After asking 'Does aerobic respiration fully explain the bubbles and sour taste?', teacher counts silently to 12 and then says 'Talk to the person next to you before anyone answers aloud — 90 seconds.'	The 'From Ugali to ATP' flow diagram grounds the abstract biochemistry in a concrete Kenyan dietary context, making the pathway feel personal and culturally relevant. The deliberate return to the anchoring phenomenon at the end of the Explain Phase creates productive incompleteness — learners recognise that aerobic respiration alone cannot account for the sealed thermos observations, generating authentic curiosity for Lesson 3. This is a critical storyline move: explanation advances but the phenomenon is not yet fully resolved.	Teacher collects a 'ticket-out' response from each learner on a sticky note or in the journal margin: learners write one sentence completing the stem 'Aerobic respiration cannot fully explain the uji thermos because...' Teacher reads these before Lesson 3 to identify conceptual gaps and misconceptions to address at the start of the next lesson.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Cellular respiration introduction](#) Source: Khan	Learners return to the class Driving Question Board created in Lesson 1. Teacher reads aloud three questions from the DQB that relate to today's lesson (for example: 'What is the cell using as fuel?', 'Why does the body need oxygen?', 'Where inside the cell does energy come from?'). For	Teacher prepares green and orange sticky dots before the lesson. When reading DQB questions aloud, teacher deliberately reads them with genuine curiosity: 'This one says — why does the body need oxygen? Has today given us enough to answer that? What do	The DQB update ritual makes the storyline progression visible and tangible. By colour-coding answered versus partially answered questions, learners see that science is a process of building understanding incrementally. Adding new questions generated by today's	The quality and specificity of new sticky-note questions serve as formative evidence of conceptual engagement. Questions that ask about 'what happens without oxygen' or 'how the yeast makes bubbles' indicate that learners have made the crucial inferential leap needed for Lesson 3. Teacher

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	each question, teacher invites a volunteer to come forward and place a green sticky dot on the question if today's lesson has provided evidence to begin answering it, or an orange dot if the question is only partially answered. Learners then write one new 'need-to-know' question generated by today's lesson onto a fresh sticky note and add it to the DQB. Teacher facilitates a 2-minute gallery walk where learners read the new questions silently and place a small tick on any new question they share.	you think?' Teacher models adding a new question to the board: 'I am going to add my own question: if aerobic respiration produces carbon dioxide, why is the carbon dioxide in the uji thermos not enough to explain all the bubbles? Let us see if your questions overlap with mine.' Teacher photographs the updated DQB with a phone or camera to document learning progression for the unit portfolio.	evidence models how good investigations create new questions, not just answers — a core scientific practice.	notes these questions publicly: 'Several of you are asking about what happens WITHOUT oxygen — that is exactly where we are heading next lesson.'
Model Building Phase 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Individually, each learner draws or updates a simple explanatory model in their journal. The model must include: (a) a labelled cell with a mitochondrion, (b) arrows showing glucose and oxygen entering the mitochondrion, (c) arrows showing carbon dioxide, water, and ATP leaving, (d) a small sketch of the uji thermos with a question mark indicating what aerobic respiration alone cannot explain, and (e) a one-sentence written caption connecting the model to the driving question. Teacher circulates and uses targeted questioning to push model precision. Learners who finish early are invited to add the Amina/Kasarani scenario to their model — showing how reduced oxygen at altitude limits ATP production in muscle cells.	Teacher uses the following specific prompts while circulating: 'Your model shows glucose going in — where did that glucose come from before it entered the cell? Can you trace it back to the ugali?' and 'Your arrow shows energy coming out — can you label that energy with its chemical name?' and 'I see you have drawn a question mark on the thermos — can you write one word next to that question mark that names what you think is missing from this explanation?' Teacher avoids correcting model drawings directly; instead uses 'What evidence from your resource card supports that choice?' WAIT TIME: When a learner is silent after a question, teacher waits 7 seconds and then offers a sentence starter: 'I think this part of the model shows...'	Individual model building at the end of the lesson serves as both a consolidation tool and a formative artefact. The deliberate inclusion of the uji thermos with a question mark ensures that learners do not reach premature closure — they are explicitly encoding productive incompleteness into their personal model, which will be revised and extended through the remaining six lessons. This cumulative model is the primary artefact of scientific thinking across the unit.	Teacher collects three to five journals (rotating across the class) at the end of the lesson to review model quality against three criteria: (1) correct labelling of reactants and products, (2) mitochondrion identified as site, (3) acknowledgement that the thermos phenomenon is not yet fully explained. Feedback is returned at the start of Lesson 3 as a brief verbal summary to the class: 'Most models correctly showed glucose and oxygen entering — here is one beautiful example that also raised the question of what happens when there is no oxygen.'

D. TEACHER REFLECTION

After the lesson, consider: Did learners genuinely engage with the productive incompleteness at the end — did they seem curious or frustrated? If frustrated, plan a 2-minute re-anchoring at the start of Lesson 3 reminding learners that great scientists spend years with unresolved questions. Were the resource cards at the right reading level? If many pairs struggled, consider pre-teaching two or three key vocabulary words (mitochondria, ATP, substrate) at the start of Lesson 3 before the investigation begins. Did the Amina/Kasarani scenario resonate? If learners showed high engagement with athletes, consider beginning Lesson 4 with a short video clip or image of a Kenyan marathon runner to maintain contextual motivation. Review the ticket-out sticky notes and DQB new questions tonight — use these to decide which misconceptions to address explicitly in the first 3 minutes of Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed and read about the process of aerobic respiration, noting that glucose and oxygen are consumed inside the mitochondria of living cells to release energy (ATP), carbon dioxide, and water. They connected this to the energy needs of Amina the Nairobi sprinter and to the food journey from ugali to ATP in human cells.
What did I learn?	Aerobic respiration requires glucose and oxygen as reactants and produces carbon dioxide, water, and ATP (energy). It occurs primarily in the mitochondria. A continuous supply of oxygen is essential for sustained energy release in active cells. The word equation is: Glucose + Oxygen yields Carbon Dioxide + Water + Energy (ATP).
How does this explain the phenomenon?	Aerobic respiration begins to explain how living organisms release energy from food, including how a student's muscle cells use glucose from ugali during exercise. However, aerobic respiration alone does NOT fully explain the bubbling and souring inside the sealed uji thermos, because that flask has no fresh oxygen entering — something else must be happening, and that process will be investigated in Lesson 3.

LESSON 3 (80 minutes (2 × 40-minute lessons)): Caught in the Act: Investigating Anaerobic Respiration in Yeast	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out a guided fermentation investigation — yeast + glucose solution in a sealed flask with a lime-water indicator tube — to gather direct evidence that living microorganisms produce CO_2 during anaerobic respiration, then use this evidence to explain the bubbles and increasing sourness observed in the anchoring phenomenon (the sealed thermos of uji).
Knowledge	Learners will know that: (1) anaerobic respiration in yeast produces CO_2 and ethanol in the absence of oxygen; (2) CO_2 turns lime water milky, providing a chemical test for its presence; (3) the word equation for anaerobic respiration in yeast is: glucose → ethanol + carbon dioxide (+energy); (4) the same process — carried out by naturally occurring yeast in fermented uji — explains both the bubbles and the increasing acidity observed in the anchoring phenomenon.
Skills	Learners will be able to: (1) set up and carry out a controlled fermentation investigation, identifying and managing variables; (2) observe, record, and compare qualitative data (lime-water turbidity, gas production, smell) from experimental and control flasks; (3) construct a word equation for anaerobic respiration in yeast; (4) use evidence from the investigation to explain a real-world observation (uji phenomenon).
Attitudes	Learners will: (1) demonstrate integrity by recording observations accurately and honestly, including null results from the control; (2) show peace and unity by collaborating harmoniously in lab groups; (3) appreciate the significance of invisible microbial processes in everyday Kenyan food systems.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	In Lesson 1 learners encountered the puzzling uji phenomenon and generated questions. In Lesson 2 they distinguished aerobic from anaerobic respiration conceptually and learned the word equations. Now, in Lesson 3, they gather their own empirical evidence by replicating (in miniature) what the yeast inside the thermos flask are doing. By the end of this lesson learners can point directly to CO_2 production and ethanol formation as the explanation for the bubbles in the sealed thermos, advancing the class Driving Question Board significantly.
Safety Notes	⚠️ SAFETY: (1) No brewing or consumption of alcoholic products — KICD CAUTION applies. (2) Lime water (calcium hydroxide solution) is an irritant — avoid contact with eyes and skin; wash immediately with water if contact occurs; have a first-aid kit accessible. (3) Yeast-glucose solutions may produce pressure in sealed containers — use the recommended loose-stopper or tubing setup only; do not seal flasks completely. (4) Clean all glassware thoroughly after the investigation. (5) Learners with respiratory conditions should be seated away from lime-water fumes. (6) Teacher must demonstrate the setup before learners assemble their own apparatus.

B. LESSON OVERVIEW

Learners work in groups of four to carry out a guided fermentation investigation comparing: (A) an experimental flask containing yeast + glucose solution and (B) a control flask containing glucose solution only (no yeast). Both flasks are connected via rubber tubing to test tubes of lime water. Over 20–30 minutes, learners observe whether gas is produced and whether it turns lime water milky. They record observations in a structured data table, share results in plenary, construct the word equation for anaerobic respiration in yeast, and return to the anchoring phenomenon to explain — for the first time with real evidence — what is producing the bubbles inside the sealed thermos of uji. The lesson is anchored in the NGSS Science and Engineering Practice of Planning and Carrying Out Investigations and Constructing Explanations.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Phase 1 — PREDICT (10 minutes) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner individually writes a prediction in their science journal before any equipment is distributed. They respond to two prompts on the board: (1) 'If we mix yeast with glucose solution in a sealed flask, what do you think will happen to the lime water in the connected test tube — and why?' (2) 'What do you think will happen in the control flask with NO yeast?' Learners then share their predictions in pairs (Think-Pair-Share) for 2 minutes before a brief whole-class share-out. Predictions are not corrected — they are posted on a designated section of the Driving Question Board labelled 'Our Predictions: Lesson 3'.	Distribute journals. Display the two prediction prompts on the board. Say: 'Before we touch any equipment, write your own prediction. Use the sentence starter: I predict that… because…. You have 3 minutes — write in silence.' [Allow 3 minutes of individual silent writing.] Then: 'Now share your prediction with your partner. You each have 1 minute.' [Allow 2 minutes.] Cold-call 3 pairs — one who predicts lime water WILL turn milky, one who predicts it will NOT, and one who is unsure. Say exactly: 'I am not telling you who is right. We are going to find out together. Keep your prediction — you will compare it to your evidence at the end.' Pin/tape prediction slips to the DQB. [WAIT TIME: 10–15 seconds after each cold-call before prompting further.]	Think-Pair-Share with DQB anchoring. This activates prior knowledge from Lessons 1–2, makes learner thinking visible, and creates cognitive dissonance that motivates the investigation. Posting predictions publicly on the DQB creates accountability — learners will return to them in Phase 3.	Teacher circulates and reads journal predictions during individual writing time. Observable indicator: learner predictions reference either 'gas/CO_2' or 'yeast being alive/doing something' — this shows whether Lesson 2 conceptual learning has transferred. Learners who predict no difference between flasks reveal a misconception about the role of yeast that the investigation will directly address.
Phase 2 — OBSERVE & INVESTIGATE (35 minutes) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12	Groups of 4 receive a labelled materials tray and a printed Investigation Card. The teacher demonstrates the setup once at the front. Learners then assemble their own apparatus: a 250 mL conical flask (Flask A — experimental) containing 150 mL warm water (~35–38°C), 5 g glucose, and a 5 g sachet of dry yeast; and a 250 mL conical flask (Flask B — control) containing 150 mL warm water, 5 g glucose, and NO yeast. Each flask is fitted with a one-hole stopper and rubber tubing leading to a test tube of clear lime water. Learners stir each flask to dissolve the glucose, then observe both setups every 5	Before distributing materials, say: 'I am going to demonstrate the setup once. Watch carefully — I will not repeat it.' Demonstrate at the front bench: show the stopper, tubing, and lime-water test tube. Emphasise: 'The stopper sits loosely — we are NOT sealing the flask completely. This is a safety rule.' Distribute trays. Give groups 5 minutes to assemble. Circulate during assembly — check every group's setup before they begin timing. During the observation period, circulate every 5 minutes and use targeted questions: 'What are you seeing in the lime water — describe it precisely.' 'Is Flask B doing the same thing? What does	Structured Inquiry with Controlled Comparison. By having a yeast flask and a no-yeast control running simultaneously, learners directly attribute any observed changes (lime-water turbidity, bubbling) to the presence of yeast — the only variable that differs. The structured data table scaffolds systematic observation and prevents vague, impressionistic recording. The Results Summary Box requires learners to move from raw data to an interpretive claim — the first step in constructing an explanation.	Observable indicators: (1) Learner data tables contain time-stamped, specific descriptions (e.g., 'lime water changed from clear to cloudy white at 10 min') rather than vague words like 'something happened'. (2) Results Summary Box correctly identifies yeast as the differing variable and attributes CO_2 production to Flask A. (3) Groups who observe NO lime-water change in Flask B record this accurately rather than inventing a result. Teacher notes which groups are struggling to distinguish 'observation' from 'inference' — this is addressed in Phase 3.

🔍 Search ARES for similar readings	minutes for 20 minutes, recording in a structured 3-column data table: Time | Flask A (yeast + glucose) observations | Flask B (glucose only) observations. Columns include: appearance of liquid in flask, smell, presence of bubbles in flask, appearance of lime water. At the 20-minute mark, learners also gently waft and smell each flask (teacher models safe smelling technique — never inhale directly). Groups photograph their setups using the class tablet if available. At 25 minutes, each group completes a Results Summary Box: 'In Flask A we observed… In Flask B we observed… The difference between A and B tells us…'	that tell you?' [WAIT TIME: 12 seconds after each question before accepting responses.] At the 15-minute mark, cold-call one group: 'Group three — read me exactly what you wrote in your data table for Flask A at the 10-minute mark.' At 25 minutes, say: 'You have 5 minutes to complete your Results Summary Box — every member of the group must write their own.' [Circulate to check individual writing.] At 30 minutes: 'Stop writing. We will now share in plenary.'		
Phase 3 — EXPLAIN & SENSEMAKING (20 minutes) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Plenary share-out: each group nominates a spokesperson to report one key observation. The teacher scribes a class data table on the board, aggregating all group results. Learners compare the class data to their individual predictions posted at the start (DQB). They then work individually for 5 minutes to: (1) write the word equation for anaerobic respiration in yeast in their journals; (2) annotate the equation by drawing arrows to label where each product connects to their observations (CO_2 → lime water turning milky; ethanol → smell; glucose → 'fuel consumed by yeast'). The class then constructs a collective written explanation for the anchoring phenomenon: the teacher poses the key question — 'So — what IS producing those bubbles inside Achieng's sealed thermos of uji in Kisumu?' — and cold-calls 4 learners to build a full explanation sentence by sentence	Run plenary share-out: 'Each group — one observation only, your most important one. Group one, start.' Scribe class table on board. After all groups share, pause and say: 'Look at the board. Look at your prediction from the start of the lesson. Turn to your partner — 30 seconds — were you right? What surprised you?' [Allow 30 seconds.] Then: 'Now, alone, write the word equation for anaerobic respiration in yeast and annotate it. 5 minutes.' [Circulate — look for correct equation: glucose → ethanol + carbon dioxide.] After 5 minutes, display the anchoring phenomenon image (thermos/uji photo or drawing) on the board or hold up the class anchor card. Say: 'Here is Achieng and her thermos. We have been trying to explain this since Lesson 1. We now have evidence. I am going to cold-call four people to build our explanation together.' Cold-call learner 1: 'Who or what is	Claim-Evidence-Reasoning (CER) Framework applied to the anchoring phenomenon. Learners move from raw observational data → word equation annotation → a structured explanatory claim about the phenomenon. By cold-calling four learners to build the explanation collaboratively, the teacher models that scientific explanation is cumulative and evidence-based, not just a single correct answer. Connecting back to Achieng and the thermos in Kisumu ensures the abstract chemistry stays rooted in the concrete, local phenomenon.	Observable indicators: (1) Word equations in journals are correctly written as: glucose → ethanol + carbon dioxide (+ energy). (2) Annotations correctly link CO_2 to lime-water observation and ethanol to smell. (3) The class explanation correctly names yeast as the organism, anaerobic respiration as the process, and CO_2 as the bubble-producing product. (4) Cold-called learner 4 can articulate the logical link between Flask A evidence and the uji phenomenon — this is the highest-order indicator for this lesson.

	on the board. The final class explanation is written in the format: 'The bubbles in the thermos are produced by [organism], which is carrying out [process], consuming [reactant] and releasing [products]. This is the same process we observed in Flask A because…'	inside the uji doing something?' [WAIT 12 seconds.] Cold-call learner 2: 'What process are they carrying out?' [WAIT 12 seconds.] Cold-call learner 3: 'What are they consuming and what are they producing?' [WAIT 12 seconds.] Cold-call learner 4: 'How does our Flask A experiment prove this?' [WAIT 15 seconds.] Write the complete explanation on the board. Say: 'Copy this into your journals. This is our best explanation so far — it may grow as we learn more.'		
Phase 4 — DRIVING QUESTION BOARD UPDATE (8 minutes) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes (or two entries in their journal DQB section if physical sticky notes are unavailable): (1) a GREEN note — one thing from today's investigation that helps answer the driving question 'How do living organisms release energy from food?'; (2) an ORANGE note — one new question that today's investigation raised that they still cannot answer (e.g., 'Does the yeast in uji produce ethanol too? Is the sourness from ethanol or something else? What happens to the yeast when all the glucose runs out?'). Learners post/write their notes on the class DQB in the designated Lesson 3 column. The teacher reads 2–3 orange questions aloud and says which upcoming lessons will address them.	Distribute sticky notes or direct learners to their DQB journal section. Say: 'Green note: one piece of evidence from today that helps answer our driving question. Orange note: one question today's investigation raised that we still cannot answer. You have 3 minutes.' [Allow 3 minutes of individual writing.] Collect/post notes on the DQB. Read aloud 2 green notes and affirm: 'Yes — this is real evidence, from your own experiment.' Read aloud 2–3 orange notes. Say: 'These are excellent scientific questions. Here is when we will answer them: [point to Lesson 4 and 5 titles on the unit roadmap]. Keep them — they show you are thinking like scientists.' Do NOT answer the orange questions now — productive uncertainty is intentional.	Metacognitive DQB Curation. Distinguishing 'what we now know' (green) from 'what we still wonder' (orange) develops scientific epistemology — learners see that one answered question generates new questions, which is how science actually progresses. The orange notes also serve as a formative pre-assessment for Lessons 4 and 5.	Observable indicators: (1) Green notes reference specific evidence from today (lime water, CO_2, yeast, Flask A) rather than general statements copied from a textbook. (2) Orange notes show genuine intellectual curiosity about the mechanism or application — they are not simply restating what was already explained. (3) Learners who cannot generate an orange question may need additional prompting — teacher notes these individuals for follow-up.
Phase 5 — MODEL BUILDING (7 minutes) 📹 **VIDEO:** Biodiversity flourishes in	Learners open their running unit model (begun in Lesson 1 as a simple sketch of the thermos and uji). They now add a 'zoom-in' inset panel to their model showing: (1) a yeast cell (labelled) inside the uji; (2) glucose molecules entering	Say: 'Take out your unit model from Lesson 1. We are going to make it smarter using today's evidence. Add a zoom-in panel — show me what is happening at the molecular and cellular level inside the uji. Use a different colour so I	Cumulative Representational Model Revision. Using a different colour pen to add new knowledge makes the progression of understanding physically visible — learners and teacher can literally see what each lesson has added.	Observable indicators: (1) Zoom-in panel correctly shows yeast as the cellular agent, glucose as input, CO_2 and ethanol as outputs. (2) Word equation is correctly written on or beside the thermos drawing. (3) Directional arrows correctly link

Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the yeast cell (drawn as simple hexagons labelled 'glucose'); (3) CO_2 bubbles and ethanol molecules leaving the yeast cell (labelled); (4) an arrow from the yeast cell to the bubbles visible in the thermos flask, with the label 'anaerobic respiration — no oxygen needed'. Learners also annotate the outside of the thermos with the word equation. Model revisions are done in a different colour pen/pencil to show knowledge growth since Lesson 1.	can see what you have added today. You have 6 minutes.' [Circulate — look for: yeast cell present, CO_2 and ethanol labelled as products, word equation on the thermos, directional arrows.] With 1 minute remaining: 'Hold up your model. Look at how much more it shows compared to Lesson 1. This model will keep growing every lesson.' Cold-call one learner to narrate their zoom-in panel to the class in 30 seconds. Close with: 'Next lesson we will investigate what happens to respiration when we change temperature — and connect that to why uji ferments faster on a hot day in Kisumu than on a cold morning in Kericho.'	The zoom-in inset technique scaffolds the move from macroscopic observation (bubbles in thermos) to microscopic mechanism (yeast cell processes), a core disciplinary practice in life science. Narrating the model aloud consolidates understanding through verbal explanation.	the cellular zoom-in to the macroscopic thermos observation. (4) Revision colour is distinct from Lesson 1 — learner has not simply redrawn from scratch but genuinely built on the prior model. Teacher photographs 3–4 models (varied quality levels) for use in Lesson 4 as comparison anchors.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. EVIDENCE QUALITY: Did learner data tables contain specific, time-stamped observations, or were they vague? If vague, how will you scaffold data recording more tightly in Lesson 4?
2. CONTROL GROUP UNDERSTANDING: Did learners correctly use Flask B (no yeast) as a control, or did some groups treat it as irrelevant? If the control's purpose was unclear, plan a 5-minute revisit at the start of Lesson 4 using the question: 'Why did we bother setting up Flask B if we already knew yeast does something?'
3. PHENOMENON CONNECTION: When cold-called to explain the uji bubbles using today's evidence, could learners name yeast, anaerobic respiration, and CO_2 as a connected chain? If not, the CER framework may need more explicit modelling in Lesson 4.
4. DQB QUALITY: Were the orange 'still wondering' notes genuinely new questions, or were they copies of the driving question? If the latter, model the distinction between 'big question' and 'specific investigable question' explicitly.
5. MODEL REVISION: Did learners show the zoom-in from macro (thermos) to micro (yeast cell), or did they only redraw the flask? Cellular-level thinking may need reinforcement — consider providing a blank yeast cell outline template for Lesson 4's model revision.
6. SAFETY COMPLIANCE: Was the no-brewing caution upheld? Were lime-water handling protocols followed? Note any safety incidents for school records.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that Flask A (yeast + glucose) produced visible bubbles in the liquid and caused the lime water in the connected test tube to turn milky/white within 10–20 minutes. Flask B (glucose only, no yeast) produced no bubbles and the lime water remained clear. Learners also noted a faint smell (ethanol/yeasty) from Flask A but not from Flask B.
What did I learn?	Learners learned that: (1) yeast carries out anaerobic respiration in the absence of oxygen; (2) the products are carbon dioxide (CO_2) and ethanol; (3) CO_2 is the gas that turns lime water milky — this is a standard chemical test for CO_2; (4) the word equation

	is: glucose → ethanol + carbon dioxide (+ energy); (5) the yeast naturally present in fermented uji carries out this same process inside the sealed thermos, producing the bubbles Achieng observed.
How does this explain the phenomenon?	Using today's experimental evidence, learners can now explain the anchoring phenomenon: the bubbles inside Achieng's sealed thermos of uji are carbon dioxide gas produced by yeast cells carrying out anaerobic respiration. The yeast consume glucose present in the fermented maize porridge, producing CO_2 (the bubbles) and ethanol. No oxygen is required, which is why the process continues even inside a sealed thermos. The increasing sourness is not yet fully explained by today's lesson (lactic acid from bacterial fermentation will be addressed in a later lesson), giving learners a legitimate remaining question for the DQB.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Anaerobic Respiration in Animals & Plants: Products, Equations, and the Sour Truth in Our Uji	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To describe and explain anaerobic respiration in animals (lactic acid pathway) and in yeast/plants (ethanol + CO_2 pathway), write word and chemical equations for both pathways, and use universal indicator to connect fermentation acidity to the sour taste of uji and mala.
Knowledge	Learners will be able to: (a) describe anaerobic respiration in living things; (b) write word and balanced chemical equations for lactic acid fermentation (animals) and alcoholic fermentation (yeast/plants); (c) relate the respiratory quotient to the type of substrate and type of respiration in living things; (d) describe the economic importance of anaerobic respiration at home and in industry.
Skills	Learners will be able to: discuss and test the properties of fermented products (acidity) using universal indicator; calculate and interpret the Respiratory Quotient (RQ) for anaerobic respiration; construct and compare annotated word and chemical equations for both anaerobic pathways; present findings in plenary using evidence-based reasoning.
Attitudes	Learners will appreciate the significance of respiration in living things, showing integrity when recording pH observations and peace when working harmoniously in groups during the fermentation acidity test.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	In Lessons 1–3 learners established that something living is at work inside the sealed thermos, identified yeast as the agent, and collected evidence that gas (CO_2) is produced. Lesson 4 is the formal EXPLAIN lesson: learners now name both anaerobic pathways, write their equations, measure the pH drop caused by the acid products, and connect the chemistry directly to the sour taste of uji and mala — answering the student in Kisumu's original question: 'What is producing the sour taste?' This lesson also introduces the lactic acid pathway in animal muscle cells, linking to the burning thigh sensation of a Nairobi sprinter, and sets the foundation for Lesson 5's exploration of economic importance and RQ calculations.
Safety Notes	Universal indicator solution and dilute ethanol-based fermented samples are mildly irritating; learners must avoid contact with eyes and wash hands after handling. No brewing or consumption of alcoholic substances (CAUTION as per KICD design). Broken glassware must be reported immediately. Teacher to keep a first-aid kit accessible on the demonstration bench.

B. LESSON OVERVIEW

Lesson 4 is the formal sensemaking (EXPLAIN) lesson in the 8-lesson evidence-gathering sequence. Learners begin by predicting what chemical products are made inside the sealed thermos of uji, then observe two demonstrations/practicals: (1) a universal indicator pH test on fresh uji vs. fermented uji vs. mala, and (2) a teacher-led limewater test confirming CO_2 from a live yeast-sugar suspension. In the Explain phase, learners construct word and balanced chemical equations for lactic acid fermentation (animals) and alcoholic fermentation (yeast/plants), explicitly connecting each product to observable evidence from the phenomenon. The DQB is updated to record newly answered questions and new ones raised (e.g., "What controls how fast fermentation happens?"). The lesson closes with a cumulative model revision in which learners annotate their working model of the thermos to show the two anaerobic pathways at the molecular level.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — PREDICT (Activate Prior Knowledge & Generate Hypotheses) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three sealed, labelled cups: Cup A (fresh uji wa kawaida — unfermented maize porridge), Cup B (fermented uji from the phenomenon thermos, prepared the day before by the teacher), Cup C (mala — fermented milk, widely available in Kenyan markets). Without opening them, learners individually write in their science journals: (1) Rank the three cups from least sour to most sour — explain your ranking. (2) What chemical substance do you think is responsible for the sour taste in Cup B and Cup C? Name it if you can. (3) The Kisumu student's thermos was sealed. Where did the 'sour-making substance' come from if nothing new was added? Learners then share predictions with a shoulder partner (Think-Pair-Share, 3 minutes).	Teacher displays the three cups on the demonstration bench and says: 'Before I let you open or smell anything, I want your brain's best prediction — not a guess, a reasoned prediction based on everything we have built so far in this unit.' [WAIT 2 minutes for individual writing.] Teacher then cold-calls 4 learners (aim for gender balance and mix of confidence levels) to share one prediction each: 'Akinyi, what substance did you name?' / 'Otieno, why did you rank mala as most sour?' / 'Wanjiru, where do you say the sour substance came from if the thermos was sealed?' / 'Kipchoge, do you agree or disagree with Wanjiru — and why?' Teacher scribes key predicted substances on the board in a T-chart: LEFT column = 'Predicted product', RIGHT column = 'Evidence we will look for'. Do NOT confirm or deny any prediction. Say: 'Hold those ideas tightly — your job this lesson is to test them against real evidence.'	Think-Pair-Share with Prediction Journaling. This strategy activates schema from Lessons 1–3 (yeast as agent, CO_2 as one product) and surfaces prior conceptions about acids and fermentation. By committing predictions to writing before evidence is gathered, learners develop a personal stake in the upcoming observations, which research shows deepens encoding of explanatory content.	Teacher circulates during the 2-minute individual writing phase and scans journals for: (a) whether learners name a specific acid (lactic acid / ethanoic acid / 'an acid') vs. only describing a sensation ('sour') — this reveals conceptual readiness; (b) whether learners reference energy release or yeast from prior lessons — this reveals storyline coherence. Observable indicator of readiness to proceed: at least 3 of the 4 cold-called learners can articulate that a chemical product (not just 'heat' or 'gas') causes the sourness.
Phase 2 — OBSERVE (Evidence Gathering: pH Testing & CO_2 Confirmation) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	ACTIVITY 1 — Universal Indicator pH Test (groups of 4, ~18 minutes): Each group receives: 3 small beakers (Cup A fresh uji, Cup B fermented uji, Cup C mala), universal indicator solution (5 drops per beaker), a pH colour chart, a white tile for colour matching, and a results table. Learners add 5 drops of universal indicator to each beaker, observe colour change, match to pH chart, record pH value, and note observations (colour, smell if teacher permits a brief 'sniff test' at arms length). Learners also test a	Before Activity 1, teacher models the universal indicator procedure at the demonstration bench: 'I am adding exactly 5 drops — count with me. I am matching against the pH chart in good light against the white tile.' Distribute materials and say: 'You have 15 minutes. Recorder writes, Observer matches colour, Reporter will share results in plenary, Safety Monitor checks no splashing.' [Circulate — visit each group once in the first 5 minutes to check technique.] After 15 minutes, call groups to attention: 'Reporters, I	Data-Collection Table with Anchored Comparison. By testing fresh uji (control), fermented uji (phenomenon), and mala (real-world extension) side by side, learners construct a direct evidence base for the claim that fermentation produces acidic substances. Anchoring every data cell back to the thermos phenomenon (final column of the results table) prevents the practical from becoming an isolated 'recipe' and keeps causal reasoning central. The limewater demonstration adds the second	Teacher checks results tables during circulation for: (a) correct pH values recorded (fermented uji should read pH 3–4, mala pH 3.5–4.5, fresh uji pH 5.5–6.5, distilled water pH 7) — groups recording pH 7 for fermented uji have likely made a technique error and need immediate re-direction; (b) whether learners write a causal sentence in the 'Connection to taste' column or merely describe colour — causal sentences ('the lower pH means more H^+ ions, which cause the sour taste') signal readiness for the Explain phase; (c) during cold-call

Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	control (distilled water) and record its pH. Results are recorded in a structured table: Sample | Colour with UI | pH value | Relative acidity | Connection to taste. ACTIVITY 2 — Teacher Demonstration: Limewater + Yeast-Sugar Flask (~7 minutes): Teacher shows a conical flask containing warm water, sugar (sukari), and a small amount of dry yeast (the same mixture prepared 30 minutes before class). A delivery tube leads into a test tube of limewater. Learners observe the limewater turning milky. Teacher asks: 'We confirmed CO_2 in Lesson 3. Today I want you to notice what is NOT present in the limewater flask — oxygen. This yeast is working without oxygen. What does that tell us about the type of respiration?' [WAIT 10 seconds.] Cold-call 2 learners.	will cold-call three of you. Give me: sample name, pH value, and one sentence connecting it to the thermos.' Cold-call Reporter from Group 1 (Cup A), Group 3 (Cup B), Group 5 (Cup C). After Activity 2 demonstration, ask the class: 'The yeast is in a sealed flask, there is no oxygen — and yet it is releasing CO_2 and we just showed fermented products are acidic. Which equation from your prior learning could explain both of these observations at once?' [WAIT 15 seconds.] Do NOT answer — this question bridges directly into the Explain phase.	evidence strand (CO_2 production without O_2 input) needed to constrain the explanation toward anaerobic respiration.	plenary, listen for use of the word 'acid' or 'acidic' — if absent from all three reporters, pause and ask: 'What does a pH below 7 tell us about the type of substance?'
Phase 3 — EXPLAIN (Constructing Equations & Connecting Pathways to the Phenomenon) 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration	PART A — Lactic Acid Pathway in Animals (~12 minutes): Teacher introduces the context: 'Imagine Peres Jepchirchir in the final 200 metres of the Nairobi Marathon. Her muscles are working so hard that oxygen cannot be delivered fast enough. What happens?' Learners read a short structured text (1 paragraph, provided on a resource card or projected) describing lactic acid fermentation in muscle cells. Working in pairs, learners construct the word equation in their journals, then the balanced chemical equation. They annotate each molecule: what it was before respiration started, where it comes from, and where it goes. They compare with an adjacent pair and reconcile any	For Part A, teacher says: 'Before you write any equation, I want you to identify your inputs and outputs — circle them in the resource text.' [WAIT 2 minutes for annotation.] Cold-call one pair to read their word equation aloud. If correct, ask: 'Now tell me — in terms of the Kisumu student's thermos, does this pathway explain what she saw? Yes or no, and justify.' [WAIT 10 seconds.] [Expected answer: No — this pathway is in animal muscles, not in yeast. Affirm partial understanding if learner says 'not directly' and redirect to Part B.] For Part B, teacher says: 'Use your pH data from Activity 1 as your first constraint. Use your limewater observation as your second	Equation Construction with Evidence Constraints (a form of Argument-Driven Inquiry). Rather than presenting the equations and asking learners to copy them, this strategy requires learners to use their own observational data (pH values, limewater result) as constraints that the equation must satisfy. This mirrors the NGSS Science and Engineering Practice of 'Constructing Explanations' — the equation is a scientific explanation that must be consistent with evidence. The comparative table then builds the Practice of 'Obtaining, Evaluating and Communicating Information' by requiring learners to organise and distinguish three related processes simultaneously.	Exit checkpoint for Part B: Teacher asks every learner to hold up their journal open to their balanced chemical equation. Teacher scans for: (a) correct molecular formulae ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$) — common error is writing CO instead of CO_2 or omitting the coefficient 2; (b) whether learners have annotated which product = bubbles and which product = pH drop — if annotations are missing, the learner has the equation but not the causal explanation; (c) in the comparison table, the ATP row is the most diagnostic — learners who write 'none' for anaerobic ATP yield have a misconception that needs correcting (anaerobic respiration does produce a small amount of ATP, just far less than

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	differences. Word equation: Glucose → Lactic acid (+ energy) Chemical equation: $C_6H_{12}O_6$ → $2C_3H_6O_3$ (+ small amount of ATP) Learners answer two written questions: (Q1) Why does Peres feel a burning sensation in her thighs during the sprint? (Q2) What happens to the lactic acid after she crosses the finish line and rests? PART B — Alcoholic Fermentation in Yeast/Plants (~12 minutes): Teacher connects back: 'Now — the yeast inside the thermos of uji. No oxygen. Same starting molecule — glucose. But the products are different. What does yeast make instead of lactic acid?' Learners examine the evidence from Activity 1 (CO_2 confirmed, pH drop confirmed) and use these as constraints to construct the word equation, then the balanced chemical equation. Word equation: Glucose → Ethanol + Carbon dioxide (+ energy) Chemical equation: $C_6H_{12}O_6$ → $2C_2H_5OH + 2CO_2$ (+ small amount of ATP) Learners annotate: which product causes the bubbles in the thermos, which product causes the pH drop / sour taste, and which product has economic value. PART C — Comparative Table (~5	constraint. Build the equation that satisfies BOTH constraints.' [WAIT 3 minutes for pair work.] Cold-call 2 pairs to write their chemical equation on the board. If equations differ, ask the class: 'Both are on the board — which one is balanced? How do you know?' [WAIT 15 seconds.] Guide learners to count atoms on each side. For Part C, say: 'The comparison table is your thinking tool — it will tell you whether you truly understand the difference between the three pathways, or whether you are just copying equations. In 5 minutes I will cold-call one person per row.' Cold-call 4 learners (one per row) to read their entry aloud. Scribe any corrections on the board.		aerobic).

	minutes): Learners complete a 4-column comparison table in journals: Feature | Aerobic respiration (Lesson 2) | Lactic acid fermentation | Alcoholic fermentation. Rows: Location in cell / Organisms / Oxygen needed? / Products / ATP yield (relative: high/low) / pH effect on surrounding fluid.			
Phase 4 — DRIVING QUESTION BOARD (DQB) Update 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon]() Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos]() 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)]() Source: CK-12 [🔍 Search ARES for similar readings]()	The class DQB (a large chart on the classroom wall, established in Lesson 1) is reviewed as a whole class. Learners receive two sticky-note colours: GREEN = 'This question is now answered by today's evidence' and YELLOW = 'This lesson raised a NEW question for me.' Working individually for 3 minutes, each learner writes at least one green note and one yellow note. Green notes are placed over previously posted questions on the DQB. Yellow notes are added to a new 'Still wondering…' section. The teacher facilitates a 5-minute whole-class gallery review, reading selected notes aloud and asking learners to vote (thumbs up/down) on whether a question is truly answered or still partially open. Expected green notes (questions answered today): 'What makes uji taste sour?' / 'What are the products of fermentation?' / 'Does the yeast use oxygen?' / 'Why does lactic acid form in muscles?' Expected yellow notes (new questions raised): 'How does temperature affect how fast fermentation happens?' / 'Is the ethanol in uji the same as in beer?' / 'How do industries control fermentation to get more	Teacher points to the DQB and says: 'In Lesson 1 we wrote questions we could not yet answer. Today you have weapons — equations, pH data, and two complete pathways. Go through the board like a scientist reviewing evidence. If today's lesson answers a question definitively, it gets a green note. If today's lesson answered part of a question but raised a deeper one, you write both.' [WAIT 3 minutes for individual sticky-note writing.] Teacher then reads 3–4 green notes aloud, pausing after each: 'Do we all agree this question is answered? What is the answer — in one sentence?' Cold-call the learner who wrote each green note to give the one-sentence answer. For yellow notes, teacher selects 2–3 and says: 'This is an excellent new question — it will anchor Lesson 5. Write it down in your journals under the heading: Questions driving our next investigation.'	Driving Question Board Revision (DQB). The DQB is a structured knowledge-tracking tool rooted in project-based learning and NGSS storyline pedagogy. By colour-coding answered vs. newly generated questions, learners externalise their growing understanding and identify the productive edge of their knowledge — the questions they cannot yet answer. This prevents the common misconception that science is 'finished' once an equation is memorised, and it maintains the motivational energy of the anchoring phenomenon across all 8 lessons.	Teacher photographs the DQB at the end of this phase (or notes which questions received green notes). Diagnostic indicator: if fewer than 3 questions about fermentation products receive green notes, the Explain phase was insufficient and the teacher should plan a brief re-teach at the start of Lesson 5. If yellow notes focus on economic applications and controlling fermentation conditions, this signals learners are ready for the Lesson 5 economic importance content. If yellow notes reveal persistent confusion about oxygen (e.g., 'Does anaerobic mean no energy at all?'), teacher should address this in the Model Building phase immediately.

	product?' / 'What is the RQ for anaerobic respiration and how is it different from aerobic?'			
Phase 5 — MODEL BUILDING (Cumulative Model Revision: Annotating the Thermos at the Molecular Level) 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative working model of the thermos phenomenon — the annotated diagram they have been building since Lesson 1 (showing the thermos, yeast cells, glucose molecules, CO_2 bubbles, and the general 'energy release' arrow from earlier lessons). Today's task: revise the model to show the anaerobic respiration pathway at the molecular level. Specific additions required: 1. Inside the yeast cell: draw glucose (label $C_6H_{12}O_6$) being converted to ethanol (C_2H_5OH) + CO_2 — with a small ATP symbol showing energy captured. 2. Add a pH arrow: show that as ethanol and CO_2 accumulate, the pH of the uji drops (annotate with the approximate pH values from their Activity 1 data). 3. Add a second panel beside the thermos: a running figure (labelled 'Nairobi sprinter's muscle cell') showing the lactic acid pathway — glucose → lactic acid + ATP — with a 'burning sensation' label. 4. Write one 'claim sentence' at the bottom of the model: 'The sour taste in the sealed thermos is caused by [product] produced when yeast respires anaerobically, converting [substrate] into [products] without using oxygen.' Learners share models with their group, receive peer feedback using the sentence frame: 'Your model explains ___ well. One thing to add or correct is ___.' Learners	Teacher distributes models (or asks learners to open their journals to the model page) and says: 'Your model from Lesson 3 showed that yeast was the agent and CO_2 was a product. That was good science. But it was incomplete — it did not show the full chemical story. Today you have the equation. Your model must now be honest about every product and every molecule.' [WAIT 1 minute for learners to re-read their previous model.] 'You have 10 minutes to revise. I will be looking for three things: the balanced equation inside the yeast cell, the pH annotation, and the claim sentence.' [Circulate — target groups who had pH data errors in Activity 1 and verify their annotation is corrected.] After peer feedback (3 minutes), cold-call 2 learners to read their claim sentence aloud. Ask the class: 'Does this claim sentence match our evidence? Is there anything it cannot yet explain?' [WAIT 15 seconds.] Close the lesson by saying: 'Your model is still incomplete — it does not yet tell us how fast fermentation happens, or what conditions control it, or how industries use this process. Those questions — your yellow sticky notes — will drive Lesson 5.'	Cumulative Model Revision (Modelling as a Science and Engineering Practice). The model is not a one-time product; it is a living scientific explanation that grows more accurate with each lesson as new evidence is incorporated. Requiring learners to add molecular-level detail today (equations, pH values, ATP) on top of the macroscopic-level detail from earlier lessons (bubbles, sour smell, yeast cells) mirrors how scientists iteratively refine models. The peer feedback protocol using a structured sentence frame promotes the NGSS practice of 'Engaging in Argument from Evidence' and develops Communication and Collaboration competencies as required by KICD.	Teacher collects (or photographs) 6 models — 2 from groups who performed well in Activity 1, 2 from groups who made pH errors, 2 randomly selected. Review against three criteria: (1) Is the balanced chemical equation ($C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$) correctly embedded inside the yeast cell diagram? (2) Is the pH annotation consistent with the group's own Activity 1 data? (3) Does the claim sentence name a specific product (ethanol and/or CO_2) as causal, rather than just saying 'chemicals'? These three criteria map directly to the KICD assessment rubric indicators: 'Ability to relate the respiratory quotient to the type of substrate and type of respiration' (knowledge), 'Discuss and test the properties of fermented products — acidity' (skills), and 'Appreciate the significance of respiration in living things' (attitude, evidenced by the claim sentence connecting chemistry to daily life).

	make final revisions.			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:\n1. EQUATIONS: Did the majority of learners independently write a balanced chemical equation for alcoholic fermentation, or did they rely on copying from the board? If copying was dominant, consider introducing a 'closed-book equation construction' task at the start of Lesson 5 as a low-stakes retrieval practice.\n2. pH DATA QUALITY: Were the pH values recorded by groups consistent with expected ranges (fermented uji pH 3–4)? If multiple groups recorded anomalous values (e.g., pH 6–7 for fermented uji), investigate whether the fermented uji sample was sufficiently mature — prepare it at least 24 hours in advance next time, and store it in a warm location (30°C) to ensure active fermentation.\n3. LACTIC ACID CONNECTION: Did learners genuinely connect the lactic acid pathway to their own embodied experience (muscle burn during exercise), or did it remain an abstract equation? If the latter, open Lesson 5 with a 30-second physical activity (jumping jacks) and ask learners to report their sensation before introducing the economic content.\n4. DQB QUALITY: Were the yellow sticky notes substantive scientific questions (about variables, mechanisms, economics) or surface-level questions (about spelling, definitions)? Substantive yellow notes signal that the phenomenon is still driving curiosity — a strong indicator of storyline coherence.\n5. KICD CAUTION COMPLIANCE: Confirm that no learner consumed any fermented ethanol samples during the practical. If any group attempted to taste the Activity 1 samples beyond the teacher-controlled 'smell test', revise the safety briefing protocol before Lesson 5.\n6. GENDER AND INCLUSION: Review your cold-call log — did you achieve gender balance across all five phases? Were learners with lower prior attainment given opportunities to answer in the less high-stakes phases (Predict, DQB) rather than only in the more exposed Explain phase?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners added universal indicator to three samples (fresh uji, fermented uji from the sealed thermos, and mala) and recorded colour changes and pH values. They observed the limewater turning milky in a sealed yeast-sugar flask with no oxygen input, confirming CO_2 production under anaerobic conditions. Fermented uji registered pH 3–4 (orange-red with universal indicator), significantly lower than fresh uji (pH 5.5–6.5, yellow-green) and distilled water control (pH 7, green).
What did I learn?	Anaerobic respiration in yeast and plants (alcoholic fermentation) converts glucose into ethanol and carbon dioxide, releasing a small amount of energy (ATP): $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. The CO_2 produces the bubbles observed in the sealed thermos; the accumulation of acidic by-products lowers the pH of the uji, producing its sour taste. In animal muscle cells under oxygen-deficient conditions (e.g., a sprinting athlete), anaerobic respiration follows the lactic acid pathway: $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3$, producing lactic acid which causes the burning sensation in muscles. Both pathways produce far less ATP than aerobic respiration and do not require oxygen.
How does this explain the phenomenon?	The sour taste and bubbles in the Kisumu student's sealed thermos of uji are explained by yeast cells carrying out alcoholic fermentation — anaerobic respiration — using glucose in the maize porridge as a substrate. Without oxygen, yeast cannot complete aerobic respiration, so it uses a shorter pathway that produces ethanol (responsible for any alcoholic notes) and CO_2 (the bubbles). The accumulation of CO_2 dissolved as carbonic acid, plus other organic acids, lowers the pH of the uji from approximately pH 6 to pH 3–4, which the tongue detects as sourness. The same principle — an acid product from anaerobic metabolism — explains why fresh uji (unfermented, no active yeast) does not sour in the same way, and why mala (fermented milk, produced by lactic acid bacteria) is similarly sour. In animal cells like those in a Nairobi sprinter's thigh muscles, a parallel but chemically distinct pathway produces lactic acid instead of ethanol, causing the characteristic burning sensation during intense exercise.

LESSON 5 (40 minutes): Respiratory Substrates and Respiratory Quotient: What Fuel Are Our Cells Really Burning?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to distinguish between the three main respiratory substrates, calculate the respiratory quotient (RQ) from given data, and relate RQ values to the type of substrate being oxidised and the type of respiration occurring, using Kenyan dietary examples.
Knowledge	Learners will know: (1) that carbohydrates, fats, and proteins can each serve as respiratory substrates; (2) that RQ = volume of CO_2 produced divided by volume of O_2 consumed; (3) that RQ for carbohydrates is 1.0, for fats approximately 0.7, and for proteins approximately 0.8; (4) that an RQ above 1.0 indicates anaerobic respiration is contributing to energy release.
Skills	Learners will be able to: extract data from provided datasets and tables; perform RQ calculations using the formula; interpret RQ values to deduce the substrate being respired; communicate reasoning using scientific vocabulary; and relate dietary choices (ugali versus nyama choma) to differences in respiratory metabolism.
Attitudes	Learners will appreciate the connection between everyday Kenyan food choices and cellular metabolic processes, developing curiosity about how invisible molecular events underlie physical performance, nutrition, and food science.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy? — This lesson contributes by showing that the type of food substrate consumed (carbohydrate-rich ugali or fat-rich nyama choma) affects the efficiency and chemistry of respiration, with direct implications for nutrition, athletics, and understanding fermentation products in Kenyan agro-processing.
Purpose in Storyline	In Lessons 3 and 4 learners established what is happening inside the sealed uji thermos — anaerobic fermentation by yeast producing CO_2 and ethanol/lactic acid. But a deeper question remains on the Driving Question Board: does it matter what the yeast (or our own muscle cells) is using as fuel? Lesson 5 answers this by introducing respiratory substrates and the RQ tool, giving learners a quantitative lens to examine the anchoring phenomenon more precisely and to connect respiration science to Kenyan diets and food industries.
Safety Notes	

B. LESSON OVERVIEW

This 40-minute lesson is a guided-inquiry EXPLAIN lesson. Learners begin by predicting which food type — maize-based ugali (carbohydrate) or nyama choma (protein/fat) — provides more efficient fuel for respiration, anchoring in the phenomenon. They then use structured data cards and print/digital resources to discover the three respiratory substrates and the concept of respiratory quotient. In the Explain phase, learners perform guided RQ calculations for glucose, fat, and protein using provided datasets, then interpret anomalous RQ values (above 1.0) as evidence of anaerobic contribution. Finally, learners update their cumulative explanatory model by annotating the uji-thermos scenario with substrate identity and RQ, advancing their answer to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are presented with two photographs on a projected slide	Display the two food photographs and ask: 'Before we look at any	Think-Pair-Share anchored to a real Kenyan dietary dilemma;	Teacher circulates during the 90-second writing period and reads 4

📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Respiratory System Organs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	or printed card: a plate of ugali with sukuma wiki (carbohydrate-dominated meal) and a plate of nyama choma with kachumbari (fat-and-protein-dominated meal). The teacher poses the question: 'A Nairobi marathon runner eats ugali the night before the race. A weightlifter at the Moi Sports Centre eats nyama choma. Both need energy from respiration — but are their cells doing exactly the same chemistry inside?' Learners write individual 2-sentence predictions in their notebooks, specifying which substrate they think each athlete is mostly burning and why. Pairs then share predictions for 60 seconds before one representative from three pairs reads their prediction aloud. Learners also revisit the anchoring phenomenon: 'The yeast in the uji thermos has been fermenting maize starch. What substrate is the yeast using, and does it matter what substrate it uses?'	data, what does your body — or yeast — actually use as fuel during respiration? Write two sentences — name the fuel you think is used and explain your reasoning.' WAIT 90 seconds in silence. Then: 'Turn to the person next to you and compare your predictions — do you agree on the fuel type?' WAIT 60 seconds. Cold-call three pairs: 'Akinyi, what did your pair predict and what is your evidence?' After each response, record key prediction terms (carbohydrate, fat, protein, glucose, starch) on the board without evaluating. Then link to the DQB: 'One of our unanswered board questions from earlier lessons was: does the type of food the yeast eats change what it produces? Today we get the tools to answer that.'	teacher uses a prediction wall on the board (three columns: Carbohydrate fuel / Fat fuel / Protein fuel) to make prior knowledge visible before instruction.	to 6 notebooks to diagnose whether learners conflate breathing with cellular fuel use, or whether they already distinguish substrate types. This informs pacing of the Observe phase. Anecdotal notes are kept mentally; misconceptions (e.g., 'the body only uses sugar') are noted for targeted questioning in the Explain phase.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Respiratory System Organs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a two-part structured resource set. Part A is a data card (teacher-prepared or from the textbook) presenting three mini-tables showing the complete aerobic oxidation equations for glucose, a representative fat (tripalmitin), and a representative protein (alanine-based approximation), with molar volumes of O_2 consumed and CO_2 produced listed. Part B is a short reading passage (150 words) defining respiratory substrate as any organic molecule that can be oxidised in the cell to release energy in the form of ATP, and listing the three main classes — carbohydrates (e.g., glucose from ugali starch), fats (e.g., fatty acids	Distribute resource cards and say: 'You have 4 minutes to read both sides of the card independently — no talking. Focus on two things: what is a respiratory substrate, and what numbers do you notice in the tables?' WAIT 4 minutes. Then: 'Now with your group, sort the food cards into the three substrate columns. You have 3 minutes. If you disagree, argue it out — I want to hear your reasoning when I come around.' WAIT 3 minutes while circulating, listening for reasoning. Prompt struggling groups: 'What macronutrient is ugali mainly made of? So which column does it belong in?' Avoid giving answers. After sorting, display the projected	Structured resource analysis (data card + short reading) followed by a tactile card-sorting categorisation task using familiar Kenyan foods — this grounds abstract biochemistry in tangible dietary context before the calculation phase begins.	Teacher observes card-sorting decisions and listens to group justifications. Correct placement of ambiguous items (e.g., mala — protein and fat, not carbohydrate; avocado — fat) reveals depth of understanding. Persistent misconceptions (e.g., placing all foods in carbohydrate column) are flagged for direct questioning during the Explain phase transition.

	from nyama choma), and proteins (e.g., amino acids, used mainly when carbohydrate and fat stores are depleted). Learners read both resources individually for 4 minutes, then in groups of three complete a sorting task: a set of 12 small cards (printed or cut out) containing food names (ugali, honey, avocado, cow milk fat, beans, fish, mandazi dough, groundnut oil, mala, cow blood, maize grain, and ghee) are sorted into three substrate columns on a shared sheet. Groups discuss which Kenyan foods are primarily carbohydrate, fat, or protein substrates. The teacher projects the data table showing O2 consumed and CO2 produced for each substrate to prepare learners for the calculation in the Explain phase.	data table: 'Now look at the numbers in this table — for glucose, how many moles of O2 are consumed and how many moles of CO2 are produced? What do you notice about the ratio?' WAIT 45 seconds for individual silent observation before taking responses.		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Respiratory System Organs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces the Respiratory Quotient (RQ) formula: RQ = volume of CO2 produced divided by volume of O2 consumed (or moles of CO2 produced divided by moles of O2 consumed). Learners copy the formula and then work through three guided calculations using data from their resource cards. Calculation 1 (Whole-class guided): Glucose — C6H12O6 + 6O2 → 6CO2 + 6H2O. Learners identify that 6 moles of CO2 are produced and 6 moles of O2 are consumed, so RQ = 6 divided by 6 = 1.0. The teacher writes this on the board and connects it: 'When the yeast in our uji thermos ferments maize starch (mainly glucose), what substrate is it using? So what RQ would we expect if it were doing aerobic respiration?' Calculation 2 (Pair	Write the RQ formula prominently: 'RQ = CO2 produced divided by O2 consumed.' Say: 'This is our new investigative tool — it tells us what fuel the cell or organism is burning, just from measuring gases. Lets try it together for glucose.' Work through Calculation 1 step-by-step on the board, asking: 'How many moles of CO2 does the equation show? How many O2? So what is the division?' WAIT 30 seconds for learners to compute before writing the answer. For Calculation 2 announce: 'With your partner, use the fat equation on your card to find the RQ. Show every step. Two minutes — go.' WAIT 3 minutes. Cold-call a pair: 'Kamau, what did you get? Walk us through your steps.' Correct errors publicly but respectfully: 'Good attempt — check your denominator, which	Gradual release of responsibility (I do — we do — you do) for RQ calculations; use of anomalous data (RQ greater than 1.0) as a cognitive hook that reconnects to the anchoring phenomenon; summary table construction consolidates the pattern.	Teacher uses mini-whiteboards or notebook spot-checks during Calculation 2 to verify arithmetic accuracy. Targeted questions during the extension reveal whether learners can reason beyond procedural calculation to interpretive understanding. Exit check: teacher asks two randomly selected learners to state the RQ for fat from memory and explain what it means — correct unprompted recall confirms consolidation.

	work, 3 minutes): Fat — tripalmitin: $2C_{51}H_{98}O_6 + 145O_2 \rightarrow 102CO_2 + 98H_2O$. Learners calculate RQ = 102 divided by 145 = approximately 0.70. The teacher circulates, prompting: 'Is the RQ for fat higher or lower than for glucose? What does that tell you about how much oxygen fat needs per unit of CO_2 it produces?' Calculation 3 (Independent, 2 minutes): Protein approximation — RQ is given as approximately 0.8 from the data card; learners write a sentence explaining what this means relative to fat and carbohydrate. Extension challenge: The teacher asks — 'What would an RQ of 1.2 mean? Could that ever happen?' Learners predict in writing before the teacher reveals that RQ above 1.0 indicates anaerobic pathways are contributing, because CO_2 is being produced in excess of what aerobic respiration alone would generate — directly linking back to the sealed uji thermos where fermentation (anaerobic) is occurring. Learners record a summary table: Substrate — Carbohydrate: RQ 1.0; Fat: RQ 0.7; Protein: RQ 0.8; Mixed anaerobic: RQ greater than 1.0.	gas is on the bottom of our formula?' For the extension: 'If you finished early, predict what an RQ above 1 would mean about the type of respiration. Write one sentence before I tell you.' WAIT 60 seconds. Reveal: 'An RQ above 1 means anaerobic respiration is happening — MORE CO_2 is being released than aerobic pathways alone can explain. Does that sound familiar? Think about the sealed thermos.' Pause for 30 seconds of silent reflection before taking hands.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Respiratory	Learners return to the class Driving Question Board displayed at the front of the room. The teacher reads aloud three sticky-note questions that were generated in earlier lessons and remain unanswered or partially answered: (1) 'Does it matter what food the yeast in uji eats?'; (2) 'Why does the body use different foods for energy?'; (3) 'How can we measure what type of respiration is happening without	Gesture to the DQB: 'Look at our board from Lesson 1. Some of these questions have been sitting there for five lessons. Today you have a new tool — RQ. Which questions can you now answer?' Give pairs 3 minutes to write sticky notes. Circulate and read notes, selecting two or three that represent common misconceptions still present for public discussion. After posting, read a new question that foreshadows Lesson 6: 'This	DQB revision as a public metacognitive ritual — making the progression of understanding visible to the whole class and validating both resolved and newly emerging questions as evidence of scientific thinking.	The quality of the 'NOW ANSWERED' sticky notes reveals whether learners can articulate RQ-based reasoning in their own words. Teacher reads all posted notes after class to identify persistent gaps for re-teaching at the start of Lesson 6.

System Organs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	seeing inside the cell?' Learners work in pairs for 3 minutes to write on sticky notes: (a) one question they can now fully answer using today's RQ knowledge, and (b) one new question that today's lesson has raised for them (e.g., 'Does a person eating only ugali have a different RQ than one eating nyama choma?', 'What happens to RQ during intense exercise?', 'Could we measure the RQ of fermentation in the uji thermos in the lab?'). Pairs post their sticky notes on the DQB under the headings 'NOW ANSWERED' and 'NEW QUESTION RAISED'. The teacher reads three of the new questions aloud, affirming scientific curiosity and noting which will be addressed in Lessons 6, 7, and 8.	group asks — does the rate of fermentation in uji change with temperature? Excellent — that is exactly where we go next lesson.' Use affirming but non-evaluative language: 'That is a precise scientific question — what measurement would you need to answer it?'		
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Respiratory System Organs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner spends the final 5 minutes updating their individual cumulative explanatory model diagram — a diagram they have been developing since Lesson 1 to explain what is happening in the sealed uji thermos. In this update, learners must add three new annotations: (1) Label the substrate being used by the yeast — identify it as a carbohydrate (glucose from maize starch) and write its RQ value for aerobic conditions (1.0) and note that the actual uji RQ is likely above 1.0 because fermentation is anaerobic, producing extra CO2. (2) Add a small key showing the three substrates and their RQ values. (3) Write one sentence connecting the uji scenario to the driving question: 'The yeast in the sealed thermos uses glucose (a carbohydrate substrate, RQ = 1.0 aerobically) but because the flask	Say: 'You have 5 minutes to update your explanatory model of the uji thermos. You must add the substrate label, the RQ key table, and one connecting sentence. This model is your personal scientific explanation — make it precise.' WAIT 5 minutes without interrupting except to support learners who are stuck: 'Start with the substrate — what is maize porridge mostly made of? Now what is the RQ for that substrate aerobically? Now think — is the yeast in the sealed flask doing aerobic or anaerobic respiration? What does that do to the RQ?' Collect or photograph two or three models using a phone or tablet (with learner permission) to display as anchor examples at the start of Lesson 6. Close the lesson: 'Your models now show substrate, RQ, and why the uji thermos produces so much CO2. Next lesson we ask	Cumulative model annotation as a synthesis task — forces integration of new RQ knowledge with prior lessons on aerobic and anaerobic pathways; the annotated sentence functions as a claim-evidence-reasoning structure connecting substrate identity to observable phenomenon.	Teacher scans 5 to 8 models during the activity, checking for: correct substrate identification (carbohydrate/glucose), correct RQ value for aerobic conditions (1.0), and a plausible explanation for why the anaerobic uji RQ exceeds 1.0. Common error to watch for: learners who write RQ = 1.0 for the uji scenario without accounting for anaerobic contribution — this is addressed at the opening of Lesson 6.

	is sealed and oxygen is limited, anaerobic fermentation raises the effective RQ above 1.0, explaining the excess CO_2 bubbles we observe.' Learners who finish early may annotate a second panel showing how a different substrate (e.g., fat from nyama choma) would change the RQ and therefore the gas ratios produced.	— what would make the yeast respire even faster? Temperature? Sugar concentration? We will investigate that and connect it back to why uji ferments quicker on a hot Mombasa afternoon than in cool Kericho.'		

D. TEACHER REFLECTION

After the lesson, consider: Did learners successfully perform all three RQ calculations or did the fat equation (with large coefficients) cause arithmetic difficulty — if so, provide a calculator policy statement for Lesson 8. Were learners able to move from procedural RQ calculation to interpretive reasoning (RQ above 1.0 means anaerobic)? Did the card-sorting activity with Kenyan foods reveal unexpected gaps in knowledge of macronutrient composition — e.g., did learners misclassify mala or avocado? Review the DQB sticky notes after class: do new questions suggest learners are developing productive lines of inquiry toward Lessons 6 and 7, or are questions very basic (e.g., still asking what respiration is)? If the latter, plan a 3-minute recap at the start of Lesson 6. Did the cumulative model annotations show that learners can connect RQ to the anchoring phenomenon, or were annotations superficial? Strong models should be photographed and displayed anonymously as exemplars. Consider whether any learner requires differentiated support for the arithmetic component of RQ — prepare a scaffold card with the formula and a worked example for Lesson 8 assessment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We looked at data tables showing how much O_2 is consumed and how much CO_2 is produced when glucose, fat, and protein are completely oxidised during aerobic respiration. We also sorted familiar Kenyan foods (ugali, nyama choma, avocado, mala, mandazi dough, groundnut oil) into substrate categories.
What did I learn?	The three main respiratory substrates are carbohydrates (RQ = 1.0), fats (RQ approximately 0.7), and proteins (RQ approximately 0.8). The respiratory quotient (RQ) is calculated as the volume of CO_2 produced divided by the volume of O_2 consumed. An RQ above 1.0 indicates that anaerobic respiration is contributing to energy release, because fermentation produces CO_2 without consuming O_2.
How does this explain the phenomenon?	The yeast in the sealed uji thermos is using glucose (a carbohydrate substrate) as its respiratory fuel. Because the flask is sealed and oxygen becomes limited, the yeast shifts to anaerobic fermentation — this produces CO_2 without consuming O_2, pushing the effective RQ above 1.0 and explaining why so many bubbles accumulate inside the sealed thermos even though no new ingredients were added.

LESSON 6 (80 minutes (2 × 40-minute lessons)): Factors Affecting the Rate of Respiration: Why Does Uji Ferment Faster in Mombasa Than in Kericho?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate and explain how temperature, oxygen availability, substrate concentration, and surface area affect the rate of respiration, connecting experimental evidence to the anchoring phenomenon of uji fermenting at different rates in different climatic regions of Kenya.
Knowledge	Learners explain factors affecting respiration in living things (temperature, oxygen availability, substrate concentration, surface area) and relate these to rates of aerobic and anaerobic respiration in microorganisms and body cells.
Skills	Learners design and conduct a controlled investigation comparing the rate of CO_2 production in yeast suspensions under varied conditions; they record, tabulate, and interpret results; and they present findings to peers using scientific reasoning (NGSS SEP 3: Planning and Carrying Out Investigations; SEP 4: Analysing and Interpreting Data; SEP 6: Constructing Explanations).
Attitudes	Learners appreciate that manipulating environmental factors can enhance or suppress respiration rates, developing integrity by recording honest results and peace by collaborating harmoniously in group investigations.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	In Lessons 1–5, learners established what respiration is, wrote and balanced equations for aerobic and anaerobic respiration, calculated respiratory quotients, and confirmed that yeast produces CO_2 and acids during fermentation. Lesson 6 now asks: what controls how FAST these processes happen? This directly answers the anchoring phenomenon puzzle — the uji in Kisumu has been bubbling and souring faster on warm days than on cool ones. By systematically manipulating temperature, oxygen, substrate concentration, and surface area, learners gather the causal evidence needed to explain the Mombasa vs. Kericho fermentation rate difference, and to later (Lessons 7–8) connect this to industrial and economic applications.
Safety Notes	Learners must not taste any experimental yeast solutions. Hot water baths (≥60 °C) must be handled with tongs or heat-resistant gloves; teacher demonstrates safe pouring. Broken glassware must be reported immediately. No brewing or distillation of alcohol is permitted (KICD CAUTION). Wash hands thoroughly after handling yeast cultures. Ensure adequate ventilation in the laboratory to disperse CO_2 accumulation.

B. LESSON OVERVIEW

Learners open by revisiting the anchoring phenomenon — the sealed thermos of uji — and predict why the same uji might ferment at different rates in Mombasa (hot, humid coast) versus Kericho (cool, highland). They then conduct a structured small-group investigation manipulating one factor at a time (temperature, oxygen availability, substrate concentration, surface area using germinating vs. dry maize seeds) on live yeast suspensions, measuring CO_2 bubble output or lime-water turbidity as a proxy for respiration rate. Groups share results in a peer-presentation gallery walk, construct a shared class data table, and use the data to build causal explanations. The lesson closes with a DQB update and a revised annotated model of the uji thermos that now includes environmental arrows showing how external conditions regulate the rate of microbial respiration inside the flask.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Phase 1 — PREDICT: What Controls How Fast the Uji Changes? 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually read a two-sentence stimulus card: 'Aisha pours uji from her thermos at 8 a.m. in Kisumu — mild bubbling, slightly sour. Her cousin in Mombasa pours identical uji at the same time — vigorous bubbling, very sour already. Her other cousin in Kericho pours the same uji — barely any bubbles, almost no sourness yet.' Learners write a 3-minute silent prediction in their science journals: (1) List at least THREE factors you think might explain the difference. (2) For each factor, state whether you think MORE of it speeds up or slows down the fermentation, and WHY. Learners then share predictions with a shoulder partner for 2 minutes.	Teacher posts the three-city scenario on the board as a visual anchor. SAY: 'Before we touch any equipment today, I want your brain to commit to a prediction. Write it down — even if you're not sure. Scientists who commit to predictions first gather better evidence.' Allow 3 minutes of silent individual writing. Set a visible timer. After 2 minutes of partner share, cold-call 4 learners (aim for gender balance and regional spread in the room): 'Kamau — what was your top factor and your reasoning?' WAIT 12 seconds after each cold-call before accepting the answer. Record all predicted factors on the board in a T-chart: FACTOR | PREDICTED EFFECT. Do NOT correct predictions yet. SAY: 'We are going to let the yeast tell us who is right.'	Prediction Commitment + T-Chart Anchoring. By forcing a written prediction before observation, learners activate prior knowledge from Lessons 1–5 (enzyme activity, temperature and reaction rates from Sub-Strand 2.5), creating cognitive dissonance hooks that will make the experimental data more memorable. The T-chart on the board remains visible all lesson as a public record to revisit.	Teacher circulates during silent writing and scans journals. Observable indicator: learner has named at least 2 scientifically plausible factors (e.g., temperature, sugar concentration, oxygen) with a directional prediction. Learners who list only 1 factor or give contradictory directions receive a prompt: 'Think about what yeast uses as a raw material and what it needs from the environment to work faster.'
Phase 2 — OBSERVE: Investigating Factors Affecting Respiration Rate 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12	Learners work in groups of 4–5. Each group is assigned ONE primary factor to investigate using a yeast-glucose solution as the model system. Groups set up their experiment, collect data over 15 minutes, then rotate to observe (not redo) one other group's set-up for 5 minutes. FACTOR A — Temperature: Three identical conical flasks each contain 50 mL of yeast-glucose solution (1 g dried baker's yeast + 5 g glucose in 50 mL warm water, pre-prepared by teacher). Flask 1 sits in ice water (~5 °C), Flask 2 at room temperature (~25 °C), Flask 3 in a warm water bath (~40 °C). Each flask is sealed with a stopper and delivery tube leading into lime water. Learners record time taken	Before groups begin, teacher conducts a 3-minute live safety demonstration: show correct lime water tube setup, demonstrate how to use tongs with hot water bath, remind learners about the KICD CAUTION on no alcohol brewing. SAY: 'Your only job right now is to observe and record honestly — even if your results surprise you or contradict your prediction. That is what real scientists do.' Circulate continuously, spending approximately 3 minutes per group. Use targeted questions — do NOT give answers: 'What is your controlled variable in this experiment?', 'You said the bubbles slowed down — what does that tell you about the yeast's	Jigsaw Investigation with Cross-Group Observation. Each group becomes the class 'expert' on one factor. The rotation visit builds cross-factor literacy without requiring each group to run all four experiments. This mirrors authentic scientific division of labour (e.g., how KEMRI research teams share data) and ensures every learner has first-hand exposure to multiple data sets before the explanation phase.	Teacher checks data tables during circulation. Observable indicator: (1) Learner has recorded at least 3 time-point observations (not just final), (2) table has clearly labelled units (bubbles/min, minutes to lime water turning), (3) at least one controlled variable is identified in writing. Groups whose data tables show only qualitative terms ('lots of bubbles', 'a few') are prompted: 'Can you give me a number? Count bubbles for exactly 30 seconds and double it — that is your rate.'

🔍 Search ARES for similar readings	for lime water to turn milky (proxy for CO_2 production rate) and count bubbles per minute. FACTOR B — Oxygen Availability: Two identical yeast-glucose flasks. Flask 1 is sealed (anaerobic — stopper in place). Flask 2 is open to air (aerobic). Learners observe bubble rate and smell (sourness indicator for lactic/acetic acid accumulation) at 5-minute intervals. FACTOR C — Substrate Concentration: Three flasks with yeast solution but different glucose concentrations: 2 g/50 mL (low), 5 g/50 mL (medium), 10 g/50 mL (high). Bubble count per minute recorded at t=5, t=10, t=15 minutes. FACTOR D — Surface Area (Germinating vs. Dry Seeds): Two conical flasks fitted with CO_2 delivery tubes into lime water. Flask 1 contains 20 dry maize seeds. Flask 2 contains 20 germinating maize seeds (soaked 24 hours, pre-prepared). Learners observe lime water turbidity at 5-minute intervals and record. Learners complete a standardised data table in their journals: Factor | Level tested | Observation (bubble rate / lime water result) | Inferred respiration rate (high/medium/low).	activity?', 'If you were a tea farmer in Kericho and your uji fermented this slowly, what would you do differently?' After 15 minutes of data collection, ring a bell and direct groups to do a 5-minute silent rotation: walk to one neighbouring group's station, read their data table, and write one observation and one question in their own journal. Cold-call 3 learners after rotation: 'Wanjiku — what surprised you at the station you visited?' WAIT 15 seconds.		
Phase 3 — EXPLAIN: Building Causal Explanations from Class Data 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Each group prepares a 90-second verbal report using this scaffold written on the board: 'We investigated [FACTOR]. We changed [VARIABLE] from ___ to ___. We observed [DATA]. This tells us that when [FACTOR] increases/decreases, the rate of respiration [increases/decreases/peaks] because [MECHANISM LINKING TO ENZYME ACTIVITY / SUBSTRATE AVAILABILITY / OXYGEN PATHWAY]. This	During group presentations, teacher acts as a 'press conference moderator'. After each group's 90 seconds, SAY: 'The floor is open — one challenging question for this group from the rest of the class.' Wait 10 seconds for hands. If no hands, cold-call: 'Otieno — you visited this station. What would you ask them?' After all four presentations, construct the Summary Data Table collaboratively: point to an empty cell and cold-call a learner to fill it	Scaffolded Presentation + Collaborative Table Construction (NGSS SEP 6: Constructing Explanations). The fill-in-the-blank oral presentation scaffold forces learners to articulate causal mechanisms rather than merely describe observations. The teacher-facilitated class table publicly synthesises four independent data sets into a unified explanatory framework — modelling how scientists aggregate evidence from multiple	Written synthesis response (4-minute individual write). Observable indicator: response (a) names temperature as the key variable, (b) uses the word 'enzyme' or 'yeast activity' in a causal sentence, (c) mentions CO_2 or acid production as the measurable outcome, and (d) explicitly references the uji in the thermos. Responses meeting fewer than 3 criteria receive a written teacher prompt in the margin: 'What is inside the yeast

for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	explains the uji phenomenon because ___.' Groups present in sequence (Factors A → B → C → D). After all four presentations, teacher facilitates a whole-class construction of a shared Summary Data Table on the board, with columns: Factor | Direction of change | Effect on respiration rate | Mechanism | Link to uji phenomenon. Learners copy this into journals. Teacher then poses the synthesis question: 'Using this table, explain in full sentences why the Mombasa uji fermented faster than the Kericho uji by 8 a.m.' Learners write for 4 minutes individually before a think-pair-share.	in, then ask the class 'Do we agree?' (show of hands), then ask a learner who disagrees to explain. For the synthesis question, circulate during the 4-minute individual writing. After think-pair-share (2 minutes), cold-call 3 pairs: 'Tell me your Mombasa-vs-Kericho explanation.' Target the explanation: it must include temperature → enzyme activity → faster substrate breakdown → faster CO_2 and acid production. If a response is incomplete, SAY: 'You've got the what — now give me the why at the molecular level. What happens to yeast enzyme activity when temperature rises from 15 °C to 30 °C?'	studies. The Mombasa-Kericho synthesis question requires learners to transfer the class data to the original phenomenon, closing the evidence loop opened in Phase 1.	cell that controls how fast glucose is broken down?' for follow-up in Phase 4.
Phase 4 — DRIVING QUESTION BOARD (DQB) UPDATE: Tracking Our Growing Understanding 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class DQB is displayed (a large poster or whiteboard section maintained across all 8 lessons). Learners receive two sticky-note colours: GREEN = 'I can now answer this' and ORANGE = 'This lesson raised a new question for me.' Learners individually write and post sticky notes in 3 minutes. Teacher reads 4–5 notes aloud. Class votes by show of hands on which new orange questions are most important to carry into Lesson 7. The top 2 new questions are written in marker on the DQB. Learners then revisit the T-chart from Phase 1 (their original predictions) and tick, cross, or annotate each prediction based on the evidence gathered. SAY: 'A scientist who finds their prediction was wrong has not failed — they have learned something precise. Circle your most surprising result and write one sentence explaining why the evidence changed your thinking.'	Before sticky-note posting, SAY: 'Look at our driving question on the board — read it in your head. Then ask yourself: what part of that question can I now answer better than I could at the start of Lesson 1?' Allow 3 minutes of silent sticky-note writing. As learners post notes, teacher quickly clusters them into 3 categories on the DQB: ANSWERED | PARTLY ANSWERED | NEW QUESTIONS. Read one note from each category aloud. For NEW QUESTIONS, use 'parking lot' language: 'This is an excellent question — let's park it here and see if Lesson 7 on economic applications picks it up.' Cold-call 2 learners to share their prediction T-chart revision: 'Chebet — which prediction were you most wrong about, and why does the evidence change your thinking?' WAIT 12 seconds.	Driving Question Board Curation + Prediction Revision. The DQB is a live epistemic artefact that externalises the class's growing understanding across the entire unit. Colour-coded sticky notes make metacognitive progress visible. Prediction revision directly addresses the NGSS practice of using evidence to evaluate and revise explanations (SEP 6), and reinforces that science is an iterative, self-correcting process — a value modelled by Kenyan scientists such as Dr. Thomas Odhiambo (ICIPE) who revised pest-management models based on field evidence.	Teacher scans sticky notes during posting. Observable indicator: GREEN notes contain a specific factual claim (not just 'I understand respiration') — e.g., 'I can now explain why Mombasa uji ferments faster using temperature and enzyme activity.' ORANGE notes contain a genuine question in a question format. Learners who post blank or vague notes are individually prompted: 'Point to one thing on our Summary Data Table that surprised you and turn it into a question or a claim.'

Phase 5 — MODEL BUILDING: Revising the Uji Thermos Model 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative annotated model of the uji thermos (first drawn in Lesson 1, progressively revised in Lessons 2–5). Working individually for 6 minutes and then sharing with their group for 3 minutes, learners add a new layer to their model: ENVIRONMENTAL CONTROL ARROWS. They draw and label external arrows entering the thermos diagram showing how each factor investigated today (temperature of surroundings, O_2 availability inside the sealed flask, glucose concentration of the uji, surface area of starch/substrate particles) influences the rate of the yeast's respiration inside. Each arrow must carry a label of the form: '↑ Temperature → ↑ Enzyme activity → ↑ Rate of CO_2 and acid production.' Learners also annotate their model with a small inset box: 'Mombasa morning (~30 °C): HIGH rate' and 'Kericho morning (~15 °C): LOW rate.' Groups share models and provide one peer-improvement suggestion using the stem: 'Your model shows ___ clearly. One thing you could add to make it more complete is ___.'	Distribute learners' existing model pages (or direct them to the relevant journal page). SAY: 'Your model from Lesson 1 showed WHAT was happening inside the thermos. Today's evidence tells you HOW FAST it happens and WHY that speed changes. Add that layer now.' Circulate during individual work. Ask 3 targeted questions to different learners: 'Show me on your model where the oxygen factor comes in.', 'If you sealed this thermos in Mombasa, which arrow would be thickest — meaning most influential?', 'How does the germinating maize seed result connect to what is happening inside the uji — are there any germinating seeds in the uji?' (This last question is deliberately provocative — expected answer: no seeds, but the yeast cells are actively metabolising and have high enzyme activity analogous to germinating seeds.) After group sharing, cold-call one group to show their model on the document camera or hold it up and explain ONE arrow they added: 'Halima — pick your most important new arrow and explain it to the class in two sentences.' WAIT 10 seconds. Close by SAY: 'Your model is now a scientific tool you built from evidence. In Lesson 7, we will use this model to design products and processes — like improving uji fermentation for a Kisumu school canteen or optimising biogas production in a Rift Valley farm.'	Cumulative Model Revision (NGSS SEP 2: Developing and Using Models). Progressive model building across the lesson sequence externalises conceptual growth and makes prior-knowledge integration visible. Adding 'environmental control arrows' transforms the model from a static chemical diagram into a dynamic systems diagram — representing the organism (yeast) as embedded in and responsive to its environment. Peer feedback using the structured stem builds scientific communication skills (NGSS SEP 8: Obtaining, Evaluating and Communicating Information) while maintaining the collaborative values (Peace, Unity) mandated by KICD.	Teacher examines models during circulation. Observable indicator: model now contains (a) at least 3 labelled environmental arrows with correct directional effects, (b) a Mombasa vs. Kericho comparison annotation, (c) at least one arrow that uses enzyme activity as the mechanistic link (not just 'more heat = more reaction'). Models missing the enzyme-level mechanism receive a post-it prompt: 'What is inside the yeast that speeds up the chemical reactions? Add it to your model.' Teacher photographs 3–4 exemplar models (with permission) for display on the ARES platform resource board for Lesson 7 reference.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Which factor investigation produced the most compelling data — and which produced

messy or inconclusive results? What would you change in the set-up (e.g., yeast concentration, water temperature control, observation window) to improve reliability? (2) Were learners able to articulate the enzyme-activity mechanism linking temperature to respiration rate, or did they stay at the surface level ('heat makes things faster')? If the mechanistic language was weak, plan a 5-minute enzyme model revisit (from Sub-Strand 2.5: Rates of Reactions) at the start of Lesson 7. (3) Did the Mombasa-vs-Kericho synthesis question successfully connect experimental data back to the anchoring phenomenon? Were there learners who still could not make that connection at the end of Phase 3? Note their names for targeted follow-up. (4) Review the DQB orange sticky notes: do the new questions raised by learners align with the economic applications content planned for Lesson 7 (industrial fermentation, biogas, yoghurt production in Kenya)? If yes, use those student questions as the opening hook for Lesson 7. If learners raised questions outside the planned scope (e.g., about respiration in athletes — a context relevant to Kenyan runners like Eliud Kipchoge), consider whether a brief 3-minute acknowledgement at the start of Lesson 7 would strengthen motivation. (5) Financial literacy PCI check: did any group spontaneously connect their factor data to economic decision-making (e.g., 'a mandazi baker should keep their dough warm')? If not, plan an explicit 2-minute economic bridge at the start of Phase 3 in Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Yeast-glucose solutions at ~40 °C produced CO_2 bubbles much faster (lime water turned milky in ~4 minutes) than solutions at ~5 °C (lime water barely changed in 15 minutes). Sealed (anaerobic) flasks showed slower initial CO_2 output but increasing sourness over time. Higher glucose concentration (10 g/50 mL) produced faster and more sustained bubble rates than low concentration (2 g/50 mL). Germinating maize seeds turned lime water milky significantly faster than dry seeds of equal number, indicating higher respiration rates in metabolically active seeds.
What did I learn?	The rate of respiration in living organisms — whether yeast in uji or cells in germinating seeds — is controlled by at least four key factors: (1) Temperature: higher temperatures increase enzyme activity and therefore increase respiration rate, up to an optimum beyond which enzymes denature; (2) Oxygen availability: aerobic respiration is faster and more efficient, but anaerobic respiration proceeds when oxygen is limited, producing acids and CO_2; (3) Substrate concentration: more glucose (or other respiratory substrate) provides more fuel, increasing respiration rate up to enzyme saturation; (4) Surface area and metabolic activity: germinating seeds have greater metabolic activity and surface area for gas exchange, producing more CO_2 per unit time than dormant dry seeds. These factors explain why the uji in the Kisumu thermos ferments noticeably faster on a warm Mombasa afternoon (~30–35 °C) than during a cool Kericho highland morning (~13–16 °C) — the higher temperature accelerates yeast enzyme activity, increasing the rate of anaerobic fermentation, CO_2 production, and acid accumulation.
How does this explain the phenomenon?	The rate of anaerobic respiration (fermentation) in the yeast cells inside the sealed uji thermos is directly regulated by the temperature of the surroundings, the concentration of glucose in the uji, and the availability of oxygen diffusing through the porridge. At the warmer coastal temperatures of Mombasa, yeast enzymes catalyse the breakdown of glucose to ethanol and carbon dioxide at a faster rate — producing more bubbles and more lactic/acetic acid (sourness) per unit time. In the cooler Kericho highlands, the same enzymes work more slowly, reducing the fermentation rate and preserving the uji's freshness longer. This also explains why sealed, glucose-rich uji ferments faster than fresh unsweetened uji wa kawaida — the substrate concentration directly fuels enzyme activity. Understanding and controlling these factors has direct economic value: food processors in Nairobi can maintain optimal fermentation temperatures to standardise uji, yoghurt, and togwa production; biogas plants in the Rift Valley can maximise methane output by controlling substrate concentration and temperature; and farmers can store grain safely by keeping it cool and dry to suppress microbial respiration.

LESSON 7 (80 minutes (2 × 40-minute lessons)): DQB Revision & Economic Importance of Anaerobic Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate their understanding of the entire respiration unit by revising the Driving Question Board, then investigate and present the economic importance of anaerobic respiration — connecting fermentation science directly to Kenyan agro-processing industries, entrepreneurship, and Vision 2030 goals.
Knowledge	Learners describe the economic importance of anaerobic respiration at home and in industry, including yoghurt/mala production, uji/togwa fermentation, biogas generation, silage making, and bread/mandazi leavening, and relate each process to the biochemistry established in previous lessons.
Skills	Learners search for information on the economic importance of anaerobic respiration using digital/print media, evaluate evidence gathered across the unit, revise and annotate the DQB, and present findings clearly using structured formats (poster, oral, or digital slides).
Attitudes	Learners appreciate the significance of anaerobic respiration in creating economic value, demonstrate integrity by observing ethical behaviour when discussing value-addition of food products, and show responsibility by explicitly acknowledging the CAUTION against brewing alcoholic substances.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	By Lesson 7 learners have observed, measured, and explained the biochemistry of both aerobic and anaerobic respiration across six prior lessons. Now they step back from the microscopic scale — CO_2 bubbles, lactic acid, yeast cells — and ask: so what? They return to the sealed thermos of uji and recognise that the invisible process they have been studying for six lessons is the same engine driving a multi-billion-shilling sector of Kenya's economy. Crossing off answered questions on the DQB and surfacing remaining gaps makes the learning journey visible. Presenting on biogas in the Rift Valley, mala in Nairobi supermarkets, or silage on a Kericho dairy farm shows that the yeast bubbling in that thermos is not just a curiosity — it is a technology.
Safety Notes	CAUTION: No brewing of alcoholic substances (KICD-mandated). This must be stated explicitly during lesson introduction and revisited during presentations. Learners researching fermentation processes must be reminded that the school project context is limited to food and biogas applications only. No equipment for distillation or alcohol concentration should be present or discussed approvingly.

B. LESSON OVERVIEW

Lesson 7 is a two-part consolidation and application lesson. In Part A (≈35 min), learners conduct a structured Driving Question Board (DQB) Revision: working in home groups, they review all questions posted since Lesson 1, colour-code them (GREEN = fully answered; AMBER = partially answered; RED = still unanswered), and record the evidence from previous lessons that justifies each colour. The teacher facilitates a whole-class gallery walk and cold-call debrief to surface consensus answers and remaining gaps. In Part B (≈40 min), learner groups research and present on an assigned economic application of anaerobic respiration in Kenya — yoghurt/mala production, uji/togwa fermentation, Rift Valley dairy biogas, silage making, or mandazi/bread leavening — using digital or print media gathered in advance. Each group frames their application using the Claim–Evidence–Reasoning (CER) structure and explicitly links back to the anchoring phenomenon. A mandatory CAUTION segment on alcoholic brewing is embedded. The lesson closes with a cumulative model update in which learners add an "Economic Output" layer to the class respiration model built across the unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1 — PREDICT: What Questions Have We Actually Answered? 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a printed or hand-drawn mini-copy of the class DQB (all questions posted from Lessons 1–6). Working individually for 4 minutes, they mark each question with a personal colour-code: GREEN (I can answer this confidently with evidence), AMBER (I have partial understanding), or RED (I am still unsure). They also write one sentence of justification for their three most confident GREEN answers. Learners then share their colour-codes with a shoulder partner for 2 minutes, noting disagreements.	Distribute mini-DQB sheets as learners settle. Say: 'Before we look at our class board together, I want to know what YOU think we have learned. Colour-code every question honestly — this is not a test, it is a map of your own understanding.' Set a visible 4-minute timer. Circulate silently; note which questions show high RED rates across multiple learners — these will anchor the remaining-gaps discussion. After 4 minutes say: 'Turn to your shoulder partner. Find ONE question where you disagree on colour. Be ready to explain your evidence.' Allow 2 minutes. Do NOT resolve disagreements yet — tension drives the gallery walk.	Metacognitive Self-Assessment: By rating their own understanding before the class debrief, learners activate prior knowledge and generate genuine curiosity about unresolved questions. Disagreements between partners create productive cognitive dissonance that motivates the gallery walk.	Teacher scans mini-DQB sheets during circulation. Observable indicator: learner can write at least one evidence-based justification sentence for a GREEN question (e.g., 'We proved CO_2 is released in anaerobic respiration using the limewater experiment in Lesson 3'). Learners who cannot justify any GREEN answer are flagged for targeted support during the gallery walk.
2 — OBSERVE: DQB Gallery Walk & Evidence Audit 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook)	The class moves to the physical DQB display (wall or board). In home groups of 4–5, learners conduct a structured gallery walk (8 minutes): each group is assigned a section of the DQB and must (a) agree on a colour-code for each question in their section, (b) write the lesson number and evidence source that justifies their colour on a sticky note beside the question, and (c) identify the ONE question in their section they feel most confident answering and nominate a spokesperson. Groups rotate once (2 minutes) to peer-review the annotations of another group. The whole class then reassembles for a 10-minute cold-call debrief: the teacher selects	Assign DQB sections before the walk begins. During the walk, circulate and ask probing questions: 'What experiment gave you that evidence?', 'Can you connect this answer back to the uji thermos?' (WAIT 12 seconds after each probe before prompting). For the cold-call debrief: call on group spokespersons using a random name-picker or numbered cards — do not accept volunteer-only answers. Use the frame: 'Group [X], state your Claim, then your Evidence, then your Reasoning — 30 seconds.' After each CER response, ask ONE other learner: 'Do you agree with their reasoning? Add or challenge in 15 seconds.' For questions coded	Claim–Evidence–Reasoning (CER) Gallery Walk: This strategy makes the evidence base for scientific claims explicit and public. By requiring learners to cite lesson numbers and experiment names, it reinforces that scientific knowledge is built on evidence, not assertion — mirroring how the student in Kisumu must reason from observations about the uji, not assumptions.	Observable indicator: each group's sticky note cites a specific evidence source (lesson number, experiment name, or data). Groups that write only vague justifications ('we learned this in class') are redirected during rotation to specify the experiment. During cold-call debrief, teacher tracks which learners can correctly state the word equation or RQ value associated with their answered question.

Source: CK-12 🔍 Search ARES for similar readings	GREEN questions and asks nominated spokespersons to state the full answer using the CER frame. RED and AMBER questions are recorded on a 'Still Wondering' strip at the bottom of the DQB.	RED by most groups, say: 'Add this to our Still Wondering strip. Let's see if today's presentations answer any of these.' Explicitly state the CAUTION at this point: 'Before we move to Part B — our KICD curriculum makes one thing clear: no brewing of alcoholic substances. Any economic application we discuss today stays within food fermentation and biogas. We will talk about why this boundary exists.'		
3 — EXPLAIN: Research & Presentation — Economic Importance of Anaerobic Respiration in Kenya 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups (same home groups) are each assigned one economic application of anaerobic respiration from the following Kenya-contextualised list: (A) Yoghurt/Mala Production — lactic acid bacteria, pH drop, Kenyan dairy cooperatives (KCC, Brookside); (B) Uji/Togwa Fermentation — maize lactic acid fermentation, sourness mechanism, nutrition enhancement for children; (C) Biogas from Rift Valley Dairy Farms — methanogenic archaea, CH_4 production, replacing firewood, carbon credits; (D) Silage Making — anaerobic preservation of Napier grass and maize stover for dry-season livestock feed in Nakuru and Uasin Gishu; (E) Bread and Mandazi Leavening — yeast CO_2 production, dough rising, small-scale baking enterprises in Nairobi and Mombasa. Each group uses available digital devices or print handouts (pre-loaded on ARES offline platform) to research their application for 12 minutes. They construct a CER poster or digital slide with the following required sections: (1) What process of anaerobic respiration is involved	Before research begins, display the five application titles on the board and hand each group a structured research scaffold card with the five required sections printed as headings. Say: 'You have 12 minutes. Every section must be filled. Your final slide or poster must include the word equation for your process and ONE connection back to our uji thermos. Go.' Set a visible 12-minute timer. During research, circulate and ask targeted questions: Group A: 'What is the word equation for lactic acid fermentation? Which bacteria do this?'; Group C: 'What gas is the fuel in biogas? Where does the carbon in that gas come from originally — trace it back to maize in the uji.'; Group E: 'Why does the mandazi dough rise? What gas causes that? Is that the same gas our student in Kisumu saw bubbling in the thermos?' (WAIT 12 seconds after each question.) During presentations: cold-call one non-presenting learner after each group to restate the word equation in their own words (10-second WAIT). After all presentations, synthesise: 'Every single process we just heard about — mala,	Jigsaw Expert Presentation with CER Framing: Each group becomes the class expert on one application, ensuring all learners encounter all five contexts through peer teaching rather than teacher transmission. The CER structure forces learners to distinguish scientific explanation from economic assertion, developing critical thinking as required by the KICD General Learning Outcomes.	Observable indicators: (1) Each group's poster/slide contains a correct word equation for their anaerobic process. (2) Each group explicitly names the microorganism(s) involved. (3) Each group articulates at least one link to Kenya's economy or Vision 2030. (4) Each group makes a specific connection to the uji thermos phenomenon. Teacher uses a 4-column checklist (Word Equation ✓/✗ | Microorganism ✓/✗ | Economic Link ✓/✗ | Phenomenon Link ✓/✗) to assess each presentation in real time. Any group missing two or more columns is given a 2-minute revision opportunity before the next group presents.

	and the word equation; (2) What organisms carry out the respiration; (3) What useful product is made; (4) How this links to Kenya's economy / Vision 2030 agro-processing; (5) How this connects back to the uji thermos phenomenon. Groups present for 3 minutes each, followed by 1 minute of peer questions. Total presentation block: ≈20 minutes for all five groups.	biogas, silage, uji, mandazi — is powered by the same invisible chemistry happening in our sealed thermos. Microorganisms breaking down glucose without oxygen, releasing energy and producing useful by-products.' Restate the CAUTION explicitly: 'Notice that none of our applications involves brewing beer or chang'aa. The same biochemistry can produce alcohol, but our curriculum — and our community values — draw a clear line. We celebrate the food and energy applications.'		
4 — DRIVING QUESTION BOARD REVISION: Updating Understanding 📹 **VIDEO:** Biodiversity flourishes in Phanerozoic eon Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Anaerobic vs Aerobic Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Immediately after presentations, the whole class returns to the DQB. Each group has 2 minutes to re-examine the 'Still Wondering' RED and AMBER questions and decide: has any presentation today answered or partially answered any of these? Learners physically move questions from RED to GREEN or AMBER, or add new questions that emerged from presentations (e.g., 'How do biogas companies deal with the slurry left after methane extraction?' or 'Can the lactic acid in mala be used as a food preservative for other products?'). Learners also note on the DQB which questions remain unanswered and flag them for Lesson 8 (the final synthesising lesson). Each learner then writes a 3-sentence 'DQB Update Statement' in their exercise book: Sentence 1 — one question answered today; Sentence 2 — the evidence that answered it; Sentence 3 — one question still unanswered that they want resolved in Lesson 8.	After presentations, direct learners to the DQB: 'Groups, you have 2 minutes. Find at least one RED or AMBER question that today's presentations moved closer to GREEN. Move it. Add any new questions that came up.' Set a 2-minute timer. Then cold-call three learners at random to read their DQB Update Statement aloud. After each reading, ask the class: 'Does anyone want to add evidence to their answer or challenge their still-unanswered question?' (WAIT 10 seconds.) Use learner responses to preview Lesson 8: 'In our final lesson, we will synthesise everything — the thermos, the body, the economy — into one coherent explanation. The questions still on our Still Wondering strip will guide us.' Point to any questions about alcoholic fermentation that may have appeared and say: 'If anyone posted a question about brewing alcohol — that is a real scientific process, and we can acknowledge it exists. But our project work and community applications stop at food and biogas. That is both an ethical and a legal boundary in	DQB Revision as Visible Learning Progression: Moving question cards from RED to GREEN makes the arc of learning concrete and visible. Writing a 3-sentence Update Statement requires each learner to individually consolidate new understanding, creating a personal record of growth that can be reviewed before Lesson 8.	Observable indicator: each learner's 3-sentence DQB Update Statement correctly identifies a question answered by today's presentations AND cites a specific economic application as the evidence source. Statements that are vague or do not cite evidence are flagged. Teacher collects 5–8 books at random to review Update Statements before Lesson 8.

		Kenya.'		
5 — MODEL BUILDING: Adding the Economic Output Layer 📹 **VIDEO:** [Biodiversity flourishes in Phanerozoic eon] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Anaerobic vs Aerobic Respiration (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative class respiration model built and revised across Lessons 1–6 (a wall poster or shared diagram showing inputs/outputs of aerobic and anaerobic respiration at the cellular level). Working individually, each learner adds an 'Economic Output' annotation layer to their personal model copy using a different colour pen. This layer draws arrows FROM the anaerobic respiration products (CO_2, lactic acid, ethanol) TOWARD labelled economic products (CO_2 → mandazi dough rising; lactic acid → mala/yoghurt/uji sourness; CH_4 from methanogenic anaerobic digestion → biogas). Each learner writes a caption for their model: 'The same process that made the uji in the thermos bubble and sour is the foundation of [their assigned application] — an industry contributing to Kenya's Vision 2030 agro-processing goals.' In the final 3 minutes, two volunteers share their annotated models under a document camera or by holding them up, and the class gives 'warm' (strength) and 'wonder' (question) feedback.	Distribute personal model sheets (or direct learners to their exercise book model from previous lessons). Say: 'Use a new colour. Add arrows from the products we know — CO_2, lactic acid, ethanol, methane — to the real-world products you just researched. Label every arrow. Write your caption at the bottom.' Allow 6 minutes for annotation. Circulate and prompt: 'Where does the CO_2 in your mandazi model come from — trace it back to glucose. Which organism produced it?'; 'Your model shows lactic acid going to mala — but what happens to that lactic acid chemically to make the milk sour? Add that detail.' (WAIT 12 seconds after each prompt.) For the sharing segment, cold-call two volunteers (not self-selected). After each share, ask: 'One warm — something their model shows clearly. One wonder — a question their model raises for you.' Collect 3–5 models before learners leave to assess annotation quality. Close the lesson: 'Next lesson — Lesson 8 — we bring everything together. The sealed thermos, the marathon runner in Nairobi, the Rift Valley biogas plant, the mandazi seller in Mombasa — one unified explanation of respiration and why it matters. Come ready to synthesise.'	Cumulative Model Annotation (Economic Layer): Adding an economic output layer to the existing scientific model bridges disciplinary boundaries — it requires learners to simultaneously apply biochemical knowledge (what molecules are produced) and socioeconomic thinking (what value those molecules create). This mirrors the NGSS Science and Engineering Practice of Developing and Using Models to explain phenomena across scales.	Observable indicators: (1) Learner correctly draws arrows FROM named anaerobic respiration products TO named economic applications. (2) Caption explicitly names the microorganism, the process, and the economic context. (3) Model is internally consistent with the cellular-level model from Lessons 1–6 (no contradictions in inputs/outputs). Teacher reviews collected models for these three indicators and returns annotated feedback before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. DQB HONESTY: Were learners genuinely colour-coding their understanding, or performing confidence? If many GREEN codes lacked evidence justification, consider adding a 'Justify it or it stays AMBER' norm to the DQB protocol in future units.

2. PRESENTATION DEPTH: Did groups reach the required five sections in their CER posters/slides, or did they run out of research time? If most groups did not

complete the word equation or phenomenon link, increase the research scaffold card detail or pre-load specific ARES offline articles for each application to reduce search time.

3. CAUTION HANDLING: Was the CAUTION on alcoholic substances handled in a way that was clear, non-preachy, and educationally purposeful? Did learners understand the distinction between the biochemistry (which is real and discussable) and the project/community application boundary (which is firm)? If learners seemed confused or dismissive, plan a brief 5-minute explicit discussion at the start of Lesson 8 on why this boundary exists (health, legal, ethical dimensions in Kenya).

4. MODEL COHERENCE: Do learners' Economic Output annotations contradict anything in their earlier cellular-level model? Common error: learners draw ethanol as a product of lactic acid fermentation (confusion between the two anaerobic pathways). Review collected models before Lesson 8 and address this misconception explicitly in the synthesis lesson.

5. PHENOMENON CONNECTION: How many groups spontaneously connected their economic application back to the uji thermos without being prompted? If fewer than 3 of 5 groups did so, the anchoring phenomenon may need to be physically displayed and referenced more deliberately at the start of every lesson in future sequences.

6. REMAINING DQB QUESTIONS: What questions are still RED heading into Lesson 8? List them. Are they about aerobic respiration, the respiratory quotient, factors affecting respiration, or something unexpected? Adjust the Lesson 8 synthesis plan to address the most common remaining gaps.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners colour-coded their Driving Question Board, conducted a gallery walk with evidence citations, and listened to peer group presentations on five Kenyan economic applications of anaerobic respiration (mala/yoghurt, uji fermentation, Rift Valley biogas, silage, mandazi leavening).
What did I learn?	Anaerobic respiration is not only a cellular survival mechanism — it is the biochemical engine behind major sectors of Kenya's agro-processing economy, from Brookside dairy to Rift Valley biogas plants to mandazi sellers in Mombasa. The same CO_2 that bubbled in the sealed uji thermos causes mandazi dough to rise; the same lactic acid that soured the uji is what preserves mala and silage. Each application involves specific microorganisms carrying out anaerobic respiration under oxygen-limited conditions, producing economically valuable by-products. Brewing alcoholic substances is explicitly excluded from responsible application of this knowledge.
How does this explain the phenomenon?	Learners can now explain how anaerobic respiration products — CO_2, lactic acid, and methane — are harnessed in specific Kenyan industries, name the microorganisms involved, write the relevant word equations, and connect each application to the anchoring phenomenon of the fermenting uji thermos and to Kenya's Vision 2030 agro-processing goals.

LESSON 8 (80 minutes): Model Building & Project Showcase: Final Explanation of Respiration	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all evidence gathered across the unit to construct a final, comprehensive explanatory model of respiration, present their value-addition projects, and connect every concept — aerobic and anaerobic pathways, respiratory quotient, factors affecting respiration, and economic applications — back to the anchoring phenomenon of the sealed thermos of fermenting uji.
Knowledge	Learners explain the meaning of respiration in living things; describe aerobic and anaerobic respiration; relate respiratory quotient to substrate type and respiration type; explain factors affecting respiration; and describe the economic importance of anaerobic respiration at home and in industry.
Skills	Learners construct and annotate a cumulative explanatory model diagram; present project findings clearly and logically; evaluate peers' models using the KICD rubric; calculate and interpret RQ values; and revise scientific models in light of new evidence.
Attitudes	Learners appreciate the significance of respiration in living things; demonstrate integrity when evaluating their own and peers' work; and show appreciation for the economic and community value of anaerobic respiration processes such as yoghurt-making, uji fermentation, and biogas production.
Key Inquiry Question	How is aerobic and anaerobic respiration important in a society's economy?
Purpose in Storyline	This is the culminating lesson of the unit. Learners have spent seven lessons gathering evidence — observing fermentation bubbles, measuring CO_2 production, calculating RQ, investigating factors, and designing value-addition projects. Now they must do what scientists do: assemble all evidence into a coherent final explanation, compare their initial (Lesson 1) model to their final model to reveal conceptual growth, close every outstanding question on the Driving Question Board, and demonstrate — through their yoghurt, uji/porridge fermentation, or biogas projects — that they can use invisible cellular processes to solve real problems in communities like those in Kisumu, Kericho, and the Rift Valley.
Safety Notes	No alcoholic beverages may be brewed or presented as project outputs, in accordance with the KICD CAUTION. Yoghurt and fermented porridge samples prepared for tasting must have been made under hygienic conditions; if in doubt, display only — do not taste. Biogas models must be sealed and not ignited in the classroom. Learners with latex allergies should avoid handling rubber tubing without gloves.

B. LESSON OVERVIEW

In this 80-minute final lesson, learner groups take centre stage. The session opens with a brief 'Return to the Thermos' ritual — the teacher re-displays the anchoring phenomenon image and asks learners to silently compare what they now know to what they thought in Lesson 1. Groups then rotate through a gallery walk of three value-addition project stations (yoghurt, uji/porridge fermentation, biogas model), giving and receiving structured peer feedback using the KICD rubric. Each group formally presents their project in a 4-minute spotlight, explicitly linking their product to aerobic or anaerobic respiration, respiratory substrates, RQ, and economic value. The class then convenes for collaborative DQB completion — every sticky note question is answered aloud with evidence. The lesson closes with individual Final Model submissions: a fully annotated diagram that maps the anchoring phenomenon onto all unit concepts, followed by teacher-facilitated whole-class comparison of Lesson 1 naive models versus final models to make conceptual growth visible. Teacher reflection and formative assessment occur throughout using cold-calling, think-pair-share, and model annotation checks.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
1 — Predict Phase: Return to the Thermos 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the original anchoring phenomenon image: a sealed thermos flask of fermenting uji with visible bubbles, held by a student at a school in Kisumu. Each learner silently writes, in their science journal, a 3-sentence 'Then vs Now' statement: (1) What I thought was happening in Lesson 1; (2) What I now know is happening; (3) One question I still have. Learners then share with a partner (Think-Pair-Share) before two pairs are cold-called to read their 'Then vs Now' to the class. This activates prior knowledge, surfaces residual misconceptions, and primes learners to see the final lesson as a culmination rather than a starting point.	Teacher re-projects the anchoring phenomenon image and says: 'Before anyone speaks, I want you to go back to the very beginning. Look at this thermos of uji. In your journal, write three sentences — what you thought was happening in Lesson 1, what you now know, and one question you still carry.' Allow 4 minutes of silent writing. Then say: 'Turn to your partner. You have 90 seconds each — share your Then vs Now.' After pair discussion, say: 'I am going to cold-call two pairs. Listen carefully — you will need to add to what they say.' Cold-call Pair 1: 'Tell us your Lesson 1 thinking.' [WAIT 12 seconds.] Cold-call Pair 2: 'What is different in your Now statement?' [WAIT 12 seconds.] Record 3–4 key 'Now' statements on the board under the heading 'What We Now Know.' Do NOT correct yet — these will be validated during DQB closure. Probe any learner who still mentions 'the uji is rotting' — ask: 'Rotting and fermentation both involve microorganisms — what is the precise biochemical difference?'	Think-Pair-Share with 'Then vs Now' journaling. This metacognitive strategy makes conceptual change visible: learners articulate their prior (naive) model and contrast it explicitly with their current (evidence-based) model, mirroring the scientific practice of model revision (NGSS SEP 2 & 6). Writing before speaking ensures all learners engage, not only the most vocal.	Teacher scans journals during the 4-minute silent writing phase, looking for three observable indicators: (1) Learner correctly names yeast/lactic acid bacteria as the agents in the uji; (2) Learner references anaerobic respiration or fermentation with at least one equation or product (CO_2, lactic acid, ethanol); (3) Learner's residual question is conceptual (e.g., 'Why does RQ differ for fat vs carbohydrate?') rather than procedural. If more than 30% of learners still hold the 'rotting' misconception, the teacher inserts a 2-minute whole-class clarification before proceeding to the gallery walk.
2 — Observe Phase: Project Gallery Walk & Peer Assessment 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Three project stations are set up around the classroom or corridor: Station A — Yoghurt Making (using Kenyan maziwa lala/fermented milk tradition as context); Station B — Uji/Porridge Fermentation (connecting directly to the anchoring phenomenon; groups demonstrate pH change using red cabbage or litmus paper); Station C — Biogas Model (a sealed bottle system with	Before the gallery walk begins, say: 'You are not tourists — you are scientists peer-reviewing evidence. Use your Peer Assessment Card. Score honestly. Leave one warm and one wonder at each station.' Circulate during the walk. At Station B (uji fermentation), pause and ask any group present: 'This station connects most directly to our thermos. Point to exactly where in	Structured Peer Assessment Gallery Walk (NGSS SEP 7 — Engaging in Argument from Evidence). By assigning specific rubric criteria, the gallery walk moves beyond 'looking at posters' to disciplinary critique. Learners must locate evidence within each project for respiration type, RQ, factors, and economic value — the four pillars of the unit. The warm/wonder protocol builds a	Collect Peer Assessment Cards after the gallery walk. Check for: (1) Groups can correctly label each project as aerobic or anaerobic (or both, for multi-stage processes like biogas); (2) At least one RQ value or ratio is referenced on each card; (3) 'Wonder' questions are scientific (e.g., 'Why did you choose 37°C for yoghurt fermentation?') rather than cosmetic (e.g., 'Your poster is

for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	organic waste from the school kitchen — sukuma wiki stalks, ugali leftovers — producing gas collected in a balloon). Learner groups rotate through each station (5 minutes per station) carrying a Peer Assessment Card with four criteria drawn from the KICD rubric: (1) Correct identification of respiration type (aerobic/anaerobic); (2) Accurate RQ reference; (3) At least one factor affecting respiration discussed; (4) Clear economic/community benefit stated. Groups score each station 1–4 on each criterion and write one 'warm' (strength) and one 'wonder' (question/suggestion) on a sticky note left at the station.	your Peer Assessment Card you can score them on anaerobic respiration.' [WAIT 10 seconds.] At Station C (biogas), ask: 'The gas in this balloon — what is it? How do you know? What would you do to confirm?' [WAIT 15 seconds — expect CO_2 identification via limewater test.] After all rotations (15 minutes total), bring the class back together. Say: 'Every presenting group: read the warms and wonders left at your station. You have 2 minutes to prepare a 30-second response to the most challenging wonder you received.' This prepares groups for the spotlight presentations in Phase 3.	culture of constructive scientific discourse.	too small'). Flag any group whose card shows no RQ reference — prompt them directly before Phase 3 presentations begin.
3 — Explain Phase: Group Project Spotlight Presentations 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group delivers a 4-minute spotlight presentation of their value-addition project. The presentation must explicitly address four non-negotiable connection points (written on the board): (A) Which respiration pathway is operating and why; (B) What the RQ is for the substrate used and what it tells us; (C) At least two factors affecting respiration that the group controlled or observed during their project; (D) The economic or community benefit — framed in Kenyan context (e.g., how yoghurt-making can generate income for a Kisumu mama mboga, how biogas can replace firewood in a Kericho household, how fermented uji improves food security in a school like ours). After each presentation, the class has 1 minute for one cold-called question from a peer group. The presenting group responds. Teacher facilitates but does not answer for learners.	Introduce the spotlight by saying: 'Four minutes. Four connection points on the board — A, B, C, D. Every presentation must hit all four. I will be listening. So will your peers.' Use a visible timer. After each presentation, say: 'One question from the floor — I will cold-call.' [Spin a pen or use a random selector.] Cold-call a member of a different group: 'You heard their RQ value. Does it match the substrate they used? Explain.' [WAIT 12 seconds.] After all presentations, facilitate a 3-minute whole-class synthesis: 'What do all three projects have in common at the cellular level?' Write responses on the board. Guide toward the answer: all rely on enzyme-catalysed breakdown of organic substrates, release of ATP, and production of by-products that have economic value (lactic acid → yoghurt sour taste; CO_2 → biogas energy; CO_2 + ethanol → uji sourness and	Structured Academic Controversy with Four-Point Connection Protocol (NGSS SEP 6 — Constructing Explanations; SEP 8 — Obtaining, Evaluating, and Communicating Information). By requiring all four connection points, the protocol prevents superficial 'show-and-tell' and demands that learners construct causal explanations linking molecular biology to economic outcomes. The cold-called peer question introduces disciplinary argumentation.	During presentations, teacher uses a 4-point checklist per group: (1) Respiration pathway correctly identified with supporting equation; (2) RQ value stated and interpreted correctly; (3) Two or more factors mentioned with experimental evidence from their own project; (4) Economic benefit framed with a specific Kenyan community example. Any group missing two or more points is flagged for individual follow-up during the Model Building phase. Listen specifically for correct use of vocabulary: substrate, ATP, respiratory quotient, aerobic/anaerobic, fermentation, lactic acid, ethanol, CO_2.

		bubbles). Explicitly say: 'This is the same process happening inside the thermos of uji that confused our student in Kisumu on Day 1. Now we can explain every detail.'		
4 — Driving Question Board (DQB) Completion: Closing Every Question 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board — populated with sticky notes from all eight lessons — is displayed at the front. The teacher reads each outstanding question aloud. Learners raise hands or are cold-called to answer using evidence from the unit. For each question answered, the class votes: 'Fully answered? Partially answered? Still open?' Fully answered questions are moved to a 'We Now Know' column. Partially answered questions are annotated with a summary of what is known and what remains uncertain. The driving question itself — 'How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?' — is addressed last, with the teacher asking for a volunteer to give a full 3-sentence answer connecting all four pillars (pathway, substrate/RQ, factors, economic value). The anchoring phenomenon (sealed thermos of uji) must be named explicitly in the answer.	Display the DQB and say: 'Every question on this board was asked by someone in this room across the last eight lessons. Science works by asking and then answering. Let us close every question today.' Read the first sticky note question aloud. Say: 'Who has evidence from our lessons or projects that answers this? Do not guess — cite evidence.' [WAIT 12 seconds.] Cold-call if no hand is raised. Physically move answered notes to the 'We Now Know' column to make progress visible. When you reach the driving question, say: 'This is the question that has been guiding us since Day 1. I want a volunteer — someone who feels ready — to give a full answer in three sentences. The anchoring phenomenon — the uji thermos — must be in your answer.' After the volunteer responds [WAIT 15 seconds for the volunteer to compose], ask the class: 'What did they include? What did they leave out?' Record the final agreed answer on the board. Photograph the completed DQB for the class record. Tell learners: 'A DQB that is completely answered means the unit has done its job. Look at how many questions you have answered.'	Driving Question Board Closure (NGSS SEP 1 — Asking Questions; SEP 6 — Constructing Explanations). Physically moving sticky notes from 'questions' to 'We Now Know' makes the accumulation of scientific knowledge tangible and celebratory. Requiring explicit evidence citations prevents closure by assertion — learners must demonstrate, not merely claim, understanding. The three-sentence driving question answer synthesises the entire unit into a coherent narrative.	Observe which questions generate confident, evidenced responses and which generate hesitation or incorrect answers. If more than two questions about RQ calculation generate wrong answers, note this as a gap for teacher feedback on returned Final Models. The driving question answer is the primary summative formative indicator: a complete answer must include (1) agent — yeast/bacteria/muscle cells; (2) pathway — aerobic or anaerobic; (3) inputs/outputs — glucose, O_2, CO_2, H_2O, ATP, ethanol, lactic acid; (4) at least one economic application with a Kenyan community name.
5 — Model Building Phase: Final Annotated Explanatory	Each learner independently constructs their Final Annotated Explanatory Model — a single A4 diagram that maps the anchoring phenomenon (sealed uji thermos,	Distribute blank A4 paper and say: 'This is your Final Model. It must stand alone — someone who was not in this class should be able to read your diagram and understand	Cumulative Model Revision and Model Evolution Display (NGSS SEP 2 — Developing and Using Models; SEP 6 — Constructing Explanations). The Final Model is	Final Models are collected and assessed against the KICD rubric with four criteria: (1) Ability to relate RQ to substrate and respiration type — model shows

Model & L1 vs L8 Comparison 📹 **VIDEO:** Cellular respiration introduction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cellular Respiration (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Kisumu student, bubbles, sour smell) onto all unit concepts. The model must include and label: (a) the two respiration pathways (aerobic and anaerobic) with word or symbol equations; (b) at least two respiratory substrates with their RQ values; (c) at least three factors affecting respiration shown as arrows influencing the pathways; (d) the economic applications demonstrated in the three projects; (e) a direct annotation connecting the bubbles and sour smell in the thermos to the anaerobic fermentation products CO_2 and lactic acid/ethanol produced by yeast/lactic acid bacteria. Learners also write a 2-sentence reflection comparing their Lesson 1 model to their final model: 'In Lesson 1 I thought… Now I know… because the evidence showed…' The teacher then selects (with permission) 2 Lesson 1 models and their corresponding Final Models to display side by side on the board, facilitating a 4-minute whole-class 'Model Evolution' discussion to celebrate growth and reinforce that science is a process of progressive model revision.	how the uji thermos works, from the molecule to the market. You have 15 minutes. Work in silence — this is your individual evidence.' Circulate and provide targeted prompts (not answers): to a learner who omits factors — 'Your model shows the pathways but not what speeds them up or slows them down. Where would temperature go?'; to a learner who omits economic application — 'Your diagram ends at the cell. Where does the biogas go? Who benefits in Kericho?'; to a learner who omits the anchoring phenomenon — 'I can see the chemistry but I cannot see the thermos. Connect your model to where we started.' After 15 minutes, say: 'Add your 2-sentence reflection at the bottom. Lesson 1 thinking vs. today's thinking.' Collect all Final Models. Display two L1-vs-L8 pairs on the board. Ask the class: 'What has changed? What has been added? What has been corrected?' [WAIT 12 seconds.] Cold-call three learners. Close by saying: 'Scientists revise their models when evidence demands it. You have done exactly that over eight lessons. The student in Kisumu who was confused by her thermos of uji — you can now write her a letter explaining everything she observed. That is science.'	the capstone sensemaking artefact of the unit — it requires learners to integrate all four conceptual pillars into a single coherent visual explanation anchored in the phenomenon. The L1 vs L8 comparison makes conceptual growth concrete and visible, reinforcing that scientific understanding is built iteratively through evidence. The 2-sentence reflection is a metacognitive tool that develops 'learning to learn' — a KICD core competency.	≥2 substrates with correct RQ values and correct pathway assignment; (2) Ability to explain factors affecting respiration — ≥3 factors shown as annotated arrows with direction of effect; (3) Ability to describe economic importance of anaerobic respiration — ≥2 economic applications named with Kenyan community context; (4) Connection to anchoring phenomenon — bubbles and sour smell in the uji thermos are explicitly linked to anaerobic products in the model. The 2-sentence reflection is assessed for evidence of genuine conceptual change (not merely vocabulary acquisition). Teacher returns models with written feedback within one week.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal:

1. ANCHORING PHENOMENON CLOSURE: Did learners' Final Models convincingly connect the sealed thermos of uji — bubbles, sour smell, no external heat or ingredients — to the full biochemical explanation (anaerobic respiration by yeast/lactic acid bacteria, producing CO_2 and lactic acid/ethanol, releasing ATP)? Or did some

models still treat the phenomenon as superficial backdrop? If the latter, what earlier lesson in the sequence failed to build the connection? 2. DRIVING QUESTION BOARD: Were there questions on the DQB that remained genuinely unanswered at the end of the lesson? If yes, which concepts did they point to (RQ calculation? Aerobic vs anaerobic distinction? Economic applications?)? Plan how you will address these gaps in review or in the subsequent sub-strand (1.6 Plant Growth and Development uses similar enzymatic/cellular concepts). 3. PROJECT QUALITY AND KICD COMPLIANCE: Did all project presentations avoid brewing of alcoholic substances, in compliance with the KICD CAUTION? Were the yoghurt, uji fermentation, and biogas projects genuinely evidence-based (learners could explain the biology) or primarily product-display (learners could make the product but not explain it)? What scaffolding would you add in future to ensure the biology is not separated from the product? 4. EQUITY OF PARTICIPATION: During cold-calling and the gallery walk, which learners were consistently silent or peripheral? Were there gender patterns — for example, were female learners less likely to volunteer for the driving question final answer? Plan deliberate inclusion strategies for the next unit. 5. MODEL COMPARISON (L1 vs L8): What were the most common conceptual changes visible in the L1-vs-L8 model pairs? What misconceptions persisted even into the Final Model (e.g., confusion between breathing and cellular respiration, incorrect RQ values, treating fermentation as 'food spoiling' rather than energy release)? Use these patterns to inform your feedback on returned models and to adjust your teaching sequence for the next cohort. 6. FINANCIAL LITERACY (PCI): Did learners' project presentations genuinely engage with the economic dimension — specific Kenyan contexts, realistic income or energy estimates, community beneficiaries — or were economic benefits stated vaguely? Financial literacy is a mandated PCI for this sub-strand; ensure it is substantively addressed, not merely mentioned.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed three value-addition projects — yoghurt, uji/porridge fermentation, and a biogas model — displayed as gallery walk stations. They observed physical evidence of anaerobic respiration: pH change (colour shift of red cabbage indicator in fermented uji), gas collection in a balloon (biogas model), and texture/smell change in yoghurt. They observed two Lesson 1 naive models displayed alongside two Final Models on the board, making the visual contrast between initial and revised scientific thinking concrete.
What did I learn?	Learners consolidated that: (1) both aerobic and anaerobic respiration release ATP from organic substrates; (2) the respiratory quotient (RQ = CO_2 produced ÷ O_2 consumed) identifies substrate type — RQ = 1 for carbohydrates, < 1 for fats, > 1 for anaerobic fermentation; (3) factors such as temperature, substrate concentration, enzyme availability, and oxygen levels control the rate of both pathways; (4) anaerobic fermentation by yeast and lactic acid bacteria produces CO_2, ethanol, and lactic acid — precisely the agents responsible for the bubbles and sourness in the sealed uji thermos; (5) these same processes have direct economic value in yoghurt production, fermented porridge (uji wa kimea), and biogas generation in Kenyan households and industries.
How does this explain the phenomenon?	Learners constructed a Final Annotated Explanatory Model linking the anchoring phenomenon (sealed thermos of uji, Kisumu student, bubbles, sour smell) to all unit concepts: aerobic and anaerobic respiration pathways, respiratory substrates and RQ values, factors affecting respiration, and economic applications. They closed every outstanding question on the Driving Question Board with evidence-based answers and articulated a full 3-sentence response to the driving question: 'How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?'

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.